of
siv
va
in
wh
ge
he
pr
tic
so
pr
or
so
fir
tr

fu
th
at
tin
nu
be

w
pe
m
in
ec

w
th
2:
pe
se
sh

ON ALMOST PERIODIC FUNCTIONS AND THE THEORY OF GROUPS*

HARALD BOHR, University of Copenhagen

1. Introduction. The subject which I have chosen for my lecture, the theory of the almost periodic functions, has gradually been increased to a comprehensive and extensive theory by the contributions of numerous mathematicians in various countries. Therefore it would be a rather impossible task to try to give in a single lecture even a very cursory survey of the many different problems which have been taken up for treatment within the scope of this theory and its generalizations. The task, then, which I have set myself today is less comprehensive. First I shall try to describe quite briefly what might be called the main problem of the theory, confining myself however to the consideration of functions of a real variable, and to explain some especially important features of its solution. Subsequently I shall explain to you in a few words how this main problem could later on be fitted into a much wider class of problems than was originally the case, that is to say, that it could be considered as a problem in the so-called theory of groups. The points of view which led to this extension were first emphasized by Weyl, whereas the accomplishment of the group theoretical treatment is due to von Neumann.

2. Periodic functions. Before I begin to speak about the almost periodic functions it will be natural and convenient to say first a few words about the theory of the purely periodic functions. We shall start with a very simple, but at the same time very general notion, namely, a quite arbitrary periodic continuous motion in the plane. Let t denote the time, and let us use complex numbers $w=u+iv$ to characterize the points of the plane. Then this motion can be represented by an equation

$$w = w(t) = u(t) + iv(t),$$

where $w(t)$ is a continuous complex function for $-\infty<t<\infty$, periodic with a period p. Such a motion can of course be extremely complicated. Among these motions the most primitive one is certainly a uniform motion on a circle, for instance with its centre at the origin. Such a motion may be represented by the equation

$$w = ae^{i\lambda t}$$

where a is a complex constant and λ a real number. Let $a=re^{i\theta}$. Then r indicates the radius of the circle, and λ is the angular velocity, so that the period is $2\pi/|\lambda|$, whereas θ indicates the phase, *i.e.*, determines where on the circle the point is to be found at the time $t=0$. Such a uniform circular motion, represented by $w=ae^{i\lambda t}$, will be called a pure oscillation. For the present, we shall consider only those pure oscillations which have a given number p as one

* Rouse Ball Lecture, delivered in Cambridge, England, in May, 1946.

of their periods; let us call them mutually harmonic. If for the sake of convenience we choose $p=2\pi$, the circular motions selected in this manner are just the ones represented by $a_n e^{int}$, where n is an arbitrary integer and a_n is a complex constant. Combining an arbitrary finite number of such simple circular motions with the period 2π, *i.e.*, considering an exponential polynomial $s(t)$ of the form

$$s(t) = a_0 + a_1 e^{it} + a_{-1}e^{-it} + \cdots + a_n e^{int} + a_{-n}e^{-int},$$

we get of course again a continuous periodic motion of period 2π, which may, however, look very complicated. Such motions, produced by so-called superposition of mutually harmonic oscillations were, as is well known, not unfamiliar even to ancient Greek astronomers. Of every motion produced in this way we shall say that it can be decomposed into mutually harmonic pure oscillations. But we shall extend the meaning of this notion somewhat further, it being convenient to operate not only with finite sums, but also with infinite sums, that is to say, also to involve a limit transition. We shall here consider only a limit process uniform for all t, as the simplest possible limit transition. Thus more generally we shall say about a function $w(t)$ that it can be decomposed into mutually harmonic oscillations, if the function can be represented as the result of a uniform limit transition on finite sums of the kind in question. From the point of view of pure mathematics as well as of the applications, it is evidently a problem of decisive importance to find out which continuous periodic motions with the period 2π may in this way be composed of uniform circular motions. As is well known, this problem was solved by Weierstrass in his famous theorem that every continuous motion which is periodic with the period 2π allows such a decomposition. In other words any quite arbitrary continuous periodic motion with the period 2π can be approximated for all t by one of our polynomials $s(t)$ with an arbitrarily given degree of approximation. This being the case, the question naturally arises as to the intensity and phase with which a definite one of our oscillations e^{int} occurs in the arbitrary motion $w=w(t)$ under consideration. That is, what value does the coefficient of e^{int} assume? Evidently the coefficients of the single terms of the approximating polynomial $s(t)$ cannot be fixed exactly before we have gone to the limit, *i.e.*, as long as we only consider a polynomial $s(t)$ approximating $w(t)$ with a certain degree. However, $s_v(t)$ being a sequence of polynomials which for $v\rightarrow\infty$ converges uniformly to the given function $w(t)$, the coefficient $a_n^{(v)}$ with which the oscillation e^{int} occurs in the polynomial $s_v(t)$ will for every fixed n converge to a definite value A_n. This number A_n is called the nth Fourier coefficient of the function, and it may be said to indicate the intensity and phase with which the corresponding oscillation e^{int} occurs in the given motion $w(t)$. This nth Fourier coefficient A_n is determined by

$$A_n = \mathfrak{M}\{w(t)e^{-int}\},$$

where $\mathfrak{M}\{w(t)e^{-int}\}$ denotes the mean value $(1/2\pi)\int_0^{2\pi} w(t)e^{-int}dt$. There is another way, namely, by a more formal consideration, by which we may immediately arrive at these expressions for the Fourier coefficients. Let us write formally $w(t)$ as an infinite series

$$w(t) = \sum_{-\infty}^{\infty} A_n e^{int}$$

and make use of the fact that the system of functions e^{int} forms a normalized orthogonal system in the sense that for any two arbitrary functions $\phi(t) = e^{in_1 t}$ and $\psi(t) = e^{in_2 t}$ of the system

$$\mathfrak{M}\{\phi(t)\bar{\psi}(t)\} = \mathfrak{M}\{e^{in_1 t}e^{-in_2 t}\} = \begin{cases} 0 & \text{for} \quad n_1 \neq n_2 \\ 1 & \text{for} \quad n_1 = n_2. \end{cases}$$

Then, by multiplying our infinite series by e^{-int} and integrating term by term we get just the expression given above for the coefficient A_n of e^{int}. We have indicated above how, starting from Weierstrass' approximation theorem and by performing the limit transition, we were led to the Fourier coefficients and thus to the Fourier series of the function $w(t)$. Conversely, however, we can prove Weierstrass' approximation theorem starting from the formally formed Fourier series of the function $w(t)$. Indeed, the exponential polynomials which are directly determined by the partial sums $\sum_{-n}^{n} A_\nu e^{i\nu t}$ of the Fourier series cannot always be used as uniform approximation sums; these partial sums do, it is true, in a certain sense approximate the function best, namely in the so-called mean, but not uniformly, as we claim here; however, starting from the Fourier series we may in different ways by various so-called summation methods form finite sums $S_N(t)$ which for $N \rightarrow \infty$ converge uniformly to the given function $w(t)$. From the mere fact that it is possible from the Fourier series of the function $w(t)$ to determine, *i.e.*, to come back to the function $w(t)$, we see in particular that the function $w(t)$ is uniquely determined by its Fourier series, *i.e.*, that two different continuous periodic functions cannot have one and the same Fourier series. This fundamental theorem, the so-called uniqueness theorem, can also be formulated in the following way: The function $w(t)$ identically 0 is the only function the Fourier constants of which vanish altogether; consequently there does not exist any function $w(t)$ which may be added to the system e^{int} so that the extended system again becomes a normalized orthogonal system. This is expressed by saying that the system e^{int} is a complete normalized orthogonal system.

3. Almost periodic functions. I have dwelt at comparative length on the theory of the continuous purely periodic functions and the theory of their Fourier series, and adapted my remarks about them, in order to be able to speak all the more briefly of the corresponding more general theory of the almost periodic functions. Just as before, we start our discussion of the general theory by considering the simple pure oscillations

$$w = ae^{i\lambda t},$$

but now we include all of them in our considerations, *i.e.*, we do not select a simple mutually harmonic system by considering only those oscillations which have a given period. This gives from the very beginning an essentially different situation in view of the fact that the total number of pure oscillations has the power of the continuum whereas there exists only a denumerable number of pure oscillations with a given period. As before, so also here, we consider all finite sums of our pure oscillations, *i.e.*, all exponential polynomials of the form

$$s(t) = \sum a_n e^{i\lambda_n t}$$

where now, however, the exponents λ_n may be quite arbitrary real numbers and not all of them multiples of one and the same number as before. As formerly, we are interested in the continuous motion which is determined by the function $w=s(t)$. A principal difference, though, is that in this case the composed motion is no longer periodic; certainly the single components $a_n e^{i\lambda_n t}$ are still periodic, but in general they will have no common period, since the exponents may be incommensurable. However, as pointed out by Bohl, who in some very interesting papers studied some classes of continuous functions which include the periodic functions and are contained in the more general class of the almost periodic functions, the motion described by $w=s(t)$ must at any rate present certain periodic-like features, namely, for every $\epsilon>0$ there is an infinite number of so-called almost periods or translations numbers $\tau(\epsilon)$. By a translation number $\tau(\epsilon)$ we understand a number τ which for all t satisfies the inequality

$$|\, w(t+\tau) - w(t) \,| \leqq \epsilon.$$

In the study of the general class of motions which may be decomposed into a finite or infinite number of pure oscillations the first important problem is, of course, to find the theorem analogous to Weierstrass' theorem concerning the special case of mutually harmonic oscillations e^{int}. To put it more exactly, we ask in analogy with the former case: Which continuous motions $w=w(t)$ may be represented either by finite sums of pure oscillations or may be uniformly approached by such sums. Evidently these motions need not be periodic, but they are far from being quite arbitrary, since they must certainly present some periodic-like features. The exact solution of the problem is that the function $w(t)$ should be what I have called an almost periodic function. The definition of such a function reads: A function $f(t)$ continuous for all t is called almost periodic if, firstly, for any $\epsilon>0$ it possesses translation numbers $\tau(\epsilon)$ in the sense defined above and if furthermore, the set of translation numbers belonging to a given $\epsilon>0$ is relatively dense, which means that there do not exist arbitrarily great intervals which are free from such translation numbers $\tau(\epsilon)$.

In view of the subsequent group theoretical investigations I should like to insert a remark about another way of characterizing the almost periodic functions, a way which proves to be closely associated with the original definition given above. Already in my early investigations of almost periodic functions I had occasion to use the following theorem: Let $f(x)$ be an almost periodic function and $h_1, h_2, \cdots$ an arbitrary sequence of real numbers. Consider the sequence of functions $f(x+h_1), f(x+h_2), \cdots$ whose elements originate from the given almost periodic function $f(x)$ by the corresponding translations of the independent variable. Then we can always select a subsequence $h_{n_1}, h_{n_2}, \cdots$ so that the new sequence of functions $f(x+h_{n_1}), f(x+h_{n_2}), \cdots$ converges uniformly on the whole x-axis. Later on Bochner found the interesting result that this theorem can be converted so that we have really a new characterization of the very notion of almost periodicity. This new definition reads in exact formulation: A function $f(x)$ continuous for all x is called almost periodic, if from each sequence of functions $f(x+h_1), f(x+h_2), \cdots$ formed from $f(x)$ by translations of the x-axis a sub-sequence may be selected which converges uniformly for all x. This may also be expressed by saying that the set of functions $\{f(x+h)\}$, $-\infty < h < \infty$, formed from $f(x)$ by all possible translations is compact.

It is not very difficult to prove that every function $w(t)$ which can be decomposed into pure oscillations, *i.e.*, can be approximated uniformly by exponential polynomials $s(t)$, is an almost periodic function. The essential difficulty lies in the proof of the converse, *i.e.*, that every almost periodic motion $w(t)$ can be approximated uniformly by sums of pure oscillations. Here let me stress that the oscillations $a_n e^{i\lambda_n t}$ which occur with no quite negligible coefficients in an exponential sum $s(t)$ which approximates the given function $w(t)$ sufficiently closely must have exponents λ_n, which, in contrast to the coefficients a_n, have exactly determined values characteristic of the function $w(t)$ in question. This is intimately associated with the fact that while an oscillation $ae^{i\lambda t}$ is only slightly changed if the coefficient a is changed a little, the least change of the exponent λ means an essential change of the course of the oscillation, since we are interested in this course for all times, *i.e.*, for $-\infty < t < \infty$. In proving that an arbitrary almost periodic function $f(t)$ can really be approximated by finite sums of pure oscillations, we must therefore begin by finding a way to make the given almost periodic function $f(t)$, so to speak, deliver as its oscillation exponents certain numbers Λ_n characteristic of that function. This is obtained by connecting a Fourier series with the function, just as in the case of the periodic functions. This Fourier series, however, has here the general form

$$\sum A_n e^{i\Lambda_n t},$$

where the set of the exponents Λ_n, characteristic of the function, may be any enumerable set of real numbers and not just of integers. Thus, since the oscillation exponents Λ_n of an almost periodic function are first disclosed by its Fourier series, the Fourier series assumes, in a sense, a still more central position in the theory of the almost periodic functions than it does in the more restricted class of the purely periodic functions, where the exponents are given beforehand. In this lecture I am, of course, not able to go further into the structure of the theory, but shall only say a few words about it. The starting point is that the total non-enumerable system of all pure oscillations $e^{i\lambda t}$, where λ runs over all real numbers, forms a normalized orthogonal system, but now, of course, only if we consider the system on the whole t-axis; for, denoting by $\mathfrak{M}\{f\}$ the mean value over the infinite t-interval, namely,

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}f(t)dt,$$

we have

$$\mathfrak{M}\{e^{i\Lambda_1 t}e^{-i\Lambda_2 t}\} = \begin{cases}0 & \text{for} \quad \Lambda_1 \neq \Lambda_2,\\ 1 & \text{for} \quad \Lambda_1 = \Lambda_2.\end{cases}$$

Therefore, writing formally our given almost periodic function $f(t)$ as an infinite sum of pure oscillations, that is,

$$f(t) = \sum_1^\infty A_n e^{i\Lambda_n t},$$

we can determine the coefficient A_n of the oscillation $e^{i\Lambda_n t}$ by a formal calculation, quite analogous to that in the case of the purely periodic functions. Multiplying the equation by $e^{-i\Lambda_n t}$,and taking the mean value on both sides, we get

$$A_n = \mathfrak{M}\{f(t)e^{-i\Lambda_n t}\}.$$

However, this formal starting point must be seen from a point of view essentially different from that of the purely periodic case, where the exponents Λ_n were numbers given beforehand and where the relation was only to serve to determine the corresponding coefficients A_n. In our case where we do not know the exponents of the function but have to determine them, we proceed in the following way. For an arbitrary real λ we form the mean value

$$\mathfrak{M}\{f(t)e^{-i\lambda t}\} = a(\lambda).$$

The forming of this mean value may be interpreted as a question put to the function: Do you for the given λ contain the oscillation $e^{i\lambda t}$? The result $a(\lambda)=0$ means that the function $f(t)$ answers the question in the negative, whereas the answer $a(\lambda)\neq 0$ means that the function admits that it contains the oscillation $e^{i\lambda t}$ in question, and with a coefficient the value of which is given by the number $a(\lambda)$. Now it is relatively easy to show, and this result is a decisive point in the development of the theory, that the answer is negative for almost all values of λ, *i.e.*, that $a(\lambda)=0$ for all λ apart from an enumerable set, say $\Lambda_1, \Lambda_2, \cdots$. These values of λ are called the Fourier exponents of the function, and the corresponding mean values $A_n = a(\Lambda_n)\neq 0$ are called the Fourier coefficients of the function. With these pairs of numbers Λ_n, A_n we form the infinite series $\sum A_n e^{i\Lambda_n t}$ called the Fourier series of the almost periodic function $f(t)$ in question. Thus we have obtained a first, important, though for the present, formal, starting point of the theory.

About the further development, leading up to the final result that $f(t)$ can be approximated uniformly by finite exponential sums $s(t)$, I shall say only a few words. We have seen how to every almost periodic function $f(t)$ is attached an infinite series $\Sigma A_n e^{i\Lambda_n t}$ with real exponents Λ_n and complex coefficients A_n, the Fourier series of the function. It now becomes a question of decisive importance to ascertain whether conversely the function $f(t)$ is uniquely determined by its Fourier series, or in other words, whether two different almost periodic functions always have two different Fourier series. We may also formulate this question in another way, and ask whether our normalized orthogonal system $e^{i\lambda t}$ is complete in the set of the almost periodic functions, *i.e.*, whether we may add a further almost periodic function to this system so that the extended system becomes again a normalized orthogonal system. In a more vague formulation we may interpret the question in the following way: Can an almost periodic function really be entirely decomposed into a denumerable number of pure oscillations or does there remain an undecomposable remainder of the function after the pure oscillations given by the Fourier series have been removed? Fortunately the uniqueness theorem saying that every almost periodic function is entirely characterized by its Fourier series is valid. To prove this theorem or other theorems equivalent to it is the central, but also the most difficult point, of the theory. Several proofs exist, varying as to starting point and method. The most simple, and in a certain sense the most elementary one, is de la Vallée Poussin's proof; it may be characterized as a considerable simplification of the original and rather complicated proof of the lecturer. Other proofs were given by Norbert Wiener in connection with his interesting general spectral theory and by Hermann Weyl. Weyl's proof, based on an analogy with the theory of integral equations, has proved to be especially significant for the group-theoretical generalization of the theory about which I shall speak in a moment.

With the uniqueness theorem at our disposal, we may advance in various ways to obtain the main theorem of our theory, the theorem of the uniform approximation by exponential polynomials which is the counterpart and the generalization of Weierstrass' approximation theorem for purely periodic functions. The original proof, given by the lecturer, and generalizing a method of Bohl which was developed for an essentially more restricted class of functions, was based on a theory of Fourier series for the so-called limit-periodic functions of an infinite number of variables. Later on, Bochner succeeded in showing that the approximating exponential sums $s(t)$ in question could also be obtained directly from the Fourier series of the function $f(t)$ without the transition to functions of an infinite number of variables. Other proofs of the approximation theorem were given by Weyl and Wiener among others. Among these proofs Wiener's proof has turned out to be especially suitable for generalization. In one sense or the other all the different proofs may be regarded as summation methods, the application of which to the Fourier series of an almost periodic function produces finite sums $s(t)$ which converge uniformly to the function.

Before I proceed to place the theory within the scope of the theory of groups, I want briefly to mention that, like the classical theory of Fourier series of purely periodic functions, the theory of the Fourier series of the almost periodic functions has also been generalized in various ways by numerous mathematicians. Of special interest is a generalization due to Besicovitch, who succeeded in obtaining a class of functions almost periodic in a generalized sense, the Fourier series of which may be characterized in an especially simple way, being just the series $\Sigma A_n e^{i\Lambda_n t}$ for which $\Sigma |A_n|^2$ is convergent, while the Λ_n may be quite arbitrary real numbers.

4. The theory of almost periodic functions as a part of the theory of groups. In the remaining part of my lecture I shall try to describe briefly the points of view which made Weyl and von Neumann see the theory of the almost periodic functions (including in particular the classical theory of the purely periodic functions) in a far more general light, *i.e.*, to see it as belonging within the general theory of groups.

We may start our considerations by observing what in the present connection may be said to be the essential properties of the normalized orthogonal system in question, *i.e.*, the system of the pure oscillations $e^{i\lambda x}$. These pure oscillations may, of course, be looked upon in different ways, but their main characteristics may be said to be that they satisfy the simple functional equation

$$\phi(x + y) = \phi(x)\cdot\phi(y).$$

As is well known, this functional equation has an infinite number of solutions, both continuous and discontinuous, the latter being of a rather unpleasant or abnormal character, in fact, not even measurable in the general sense of Lebesgue. As regards the continuous solutions, they simply consist of all functions $e^{\alpha x}$, where α is an arbitrary complex constant. Selecting from among them only those for which the exponent α is a purely imaginary number $i\lambda$, we get only our pure oscillations $e^{i\lambda x}$, which thus may be characterized as the bounded continuous solutions of the functional equation in question. However, a functional equation of the type

$$\phi(x + y) = \phi(x)\cdot\phi(y)$$

is also encountered in quite another discipline, namely, in the general theory of groups, where the independent variable x, however, need not be a number, but may be a symbol of quite a different kind, whereas the values of the function $\phi(x)$ are still ordinary complex numbers. Before going into details I must first say a few words about the general notion "abstract group" which plays such a fundamental part in the mathematics of our day; this importance follows from the fact that in the theory of groups many apparently quite different investigations taken from numerous mathematical disciplines may be comprised. For the sake of simplicity I shall confine myself in what follows to the consideration of the so-called commutative or Abelian groups. Such a group is a set of a finite or an infinite set of elements $x, y, \cdots, A, B, \cdots$ (the term "element" is used if we have to deal with abstract investigations where we do not want to commit ourselves beforehand to the considered objects being of any definite kind). For these elements a single so-called composition rule is given which is called multiplication or addition, and is denoted by the multiplication sign $\cdot$ or the addition sign $+$, respectively, but which need not have anything to do with ordinary multiplication or addition, for the simple reason that the elements need not be numbers.

If we use the multiplication sign for the composition of the group, the formulas

$$A \cdot B = B \cdot A$$

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

are valid, and, furthermore, we claim that the equation

$$A \cdot X = B,$$

where A and B are two arbitrary given elements, is always satisfied by one and only one element X. If we use the addition sign, these rules read that

$$A + B = B + A$$

$$A + (B + C) = (A + B) + C$$

and that the equation

$$A + X = B$$

has one and only one solution.

Let me give you one or two examples of such Abelian groups, chosen in close connection with our subject.

First, let us consider the set of all rotations of a circle, for instance, the unit circle, about its center. Let the single rotation be characterized by its rotation angle x, where the real number x is determined modulo 2π. Then the composition rule, *i.e.*, the composition of two rotations x and y, is simply expressed by the sum $x+y$, this number being of course only determined modulo 2π. On the other hand, if we characterize the rotation by the point or complex number $X=e^{ix}$ on the unit circle, into which the point 1 is transformed by the rotation, then the composition of two rotations, determined by $X=e^{ix}$ and $Y=e^{iy}$, respectively, is expressed by ordinary multiplication $X \cdot Y=e^{i(x+y)}$.

As the second example I choose the set of the translations of a straight line into itself. Let a single translation be characterized by the (positive or negative) number x, indicating the length of the translation or, what comes to the same thing, by the point (number) x into which the origin is transformed by the translation. Then the composition of two translations given by x and y, respectively, is, of course, expressed by the ordinary sum $x+y$. Both the rotation group and the translation group are so-called topological Abelian groups, which means that, besides the structure fixed by the given composition, they have also another so-called topological structure, in consequence of which we may, for example, speak about two elements of the group lying near each other or far from each other. We observe that the first group, the rotation group, may in a certain sense be placed under the latter group, the translation group, by identifying those points on the line which differ by a multiple of 2π, or more geometrically expressed, by imagining the straight line twisted around the unit circle.

Besides these two examples of infinite groups, *i.e.*, groups containing an infinite number of elements, I shall mention a classical example of a finite Abelian group, the group of the classes of residues modulo n, where n is a positive integer. As is well known, this group is a dominating factor in an essential part of the elementary theory of numbers. Let us consider, for instance $n=3$; then the group contains only two elements which we may denote by a_1, a_2 corresponding to the two classes of integers which are relatively prime to 3. These two classes consist of the numbers of the form $3n+1$ and $3n+2$, respectively. The composition of the group is given by the following scheme

$$a_1 \cdot a_1 = a_1, \qquad a_1 \cdot a_2 = a_2 \cdot a_1 = a_2, \qquad a_2 \cdot a_2 = a_1.$$

This scheme states that the product of two numbers, both of the form $3n+1$ or both of the form $3n+2$, is a number of the form $3n+1$, whereas the product of two numbers, one of the form $3n+1$ and the other of the form $3n+2$, is of the form $3n+2$.

While generally the elements of an Abelian group are themselves not numbers, but symbols of one kind or another, they can in a natural way be connected with numbers, generally complex numbers, so that the composition rule for two arbitrary elements of the group is reproduced by ordinary multiplication of the numbers attached to these elements. This is obtained by means of the so-called group characters. A character belonging to an Abelian group is a function $\chi(X)$, where the independent variable X ranges over the elements of the group, while the values of the function are ordinary complex numbers, satisfying, for any two elements X and Y of the group, the equation

$$\chi(X \cdot Y) = \chi(X) \cdot \chi(Y)$$

where $X \cdot Y$ determines the element resulting from X and Y by the composition of the group. If the composition of the group is expressed by $X+Y$ instead of by $X \cdot Y$, the equation characteristic of a character reads

$$\chi(X + Y) = \chi(X) \cdot \chi(Y).$$

Thus we see an evident association with the functional equation valid for the exponential function, *i.e.*,

$$\phi(x + y) = \phi(x) \cdot \phi(y)$$

where x and y denote ordinary real numbers. It may be noted that it is not demanded that a character $\chi(X)$ should take two different values for two different elements X_1 and X_2; thus for any group we have the trivial so-called main character which assumes the value $\chi(X)=1$ for every X. Generally there exist both real characters for which $\chi(X)$ is a real number for every X and complex characters $\chi(X)$ which assume complex values for certain elements X. If $\chi(X)$ is a complex character, the conjugate function $\bar{\chi}(X)$ is obviously again a complex character. Further, as the functional equation provides immediately, the product $\chi_1(X)\chi_2(X)$ of two characters $\chi_1(X)$ and $\chi_2(X)$ is again a character of the group.

I shall speak briefly about the characters of the three particular Abelian groups mentioned above. As regards the group of the classes of residues modulo n with $h=\phi(n)$ elements $X_1, X_2, \cdots, X_h$, where $\phi(n)$ is the Euler function indicating the number of elements among $1, 2, \cdots, n$ which are relatively prime to n, we have at the same time, $h=\phi(n)$ different characters $\chi_1(X), \chi_2(X), \cdots, \chi_h(X)$. For these characters we have the important relations

$$\frac{1}{h}\sum_{m=1}^{h}\chi_\nu(X_m)\chi_\mu(X_n) = \begin{cases} 0 & \text{for } \nu \neq \mu, \\ 1 & \text{for } \nu = \mu \end{cases}$$

which may be said to express the fact that the characters form a normalized orthogonal system by a simple formation of mean value. It was the study of the characters of this group of the residues modulo n which was the starting point of Dirichlet's famous proof that every arithmetical progression contains an infinite number of primes.

Concerning the two other groups, the rotation group and the translation group, the elements of which we will denote by x modulo 2π and by x, respectively, where x ranges over the real numbers, we will call attention only to some of their characters, namely, the bounded characters; the unbounded characters are of no importance for our purpose. For a bounded character it is seen readily that $|\chi(x)|=1$ for all x. If furthermore we require that our bounded characters should be continuous functions on the group, (considering the group as a topological group, *i.e.*, continuous functions of the variable x), then, in the case of the rotation group (where a character must be periodic with the period 2π) these characters are only the functions e^{inx}, $n=0, \pm 1, \pm 2, \cdots$, whereas for the translation group we get the much more comprehensive set of characters $e^{i\lambda x}$, where λ is an arbitrary real number. Thus we realize that the mutually harmonic pure oscillations forming the basis of the theory of Fourier series of the purely periodic functions may be characterized from a group theoretical point of view as the bounded continuous characters of the rotation group, while the set of all pure oscillations $e^{i\lambda x}$ forming the basis of the theory of the almost periodic functions may be characterized as the bounded continuous characters of the translation group. Now we understand how the main problem solved in the theory of the purely periodic and of the almost periodic functions may, according to Weyl, be generalized to the following general problem concerning a quite arbitrary Abelian group: Which function $f(X)$ defined on the group, *i.e.*, the independent variable of which ranges over the elements of the group, can be represented by a linear composition of the bounded characters of the group? Precisely speaking, which functions defined on the given Abelian group can either be represented as a finite sum $s(X) = \sum a_\chi \chi(X)$, where the coefficients a_χ are complex constants, or can be uniformly approached by such sums?

Passing on to an outline of the solution of this general problem I start with the following remark, where for the sake of connection with the first part of my lecture it will be convenient to use an additive and not multiplicative notation for the composition of the group. We consider first a single, arbitrarily chosen bounded character $\chi(X)$ of the group. Let H be a parameter which ranges over the whole group. Then the set of all the functions $\{\chi(X+H)\}$ will, as is seen easily from the functional equation $\chi(X+H)=\chi(X)\chi(H)$, be a compact set, in the sense that from any sequence of functions $\chi(X+H_1)$, $\chi(X+H_2)$, $\cdots$ taken from the set, we can choose a subsequence of functions converging uniformly on the group. Furthermore from this property of any single bounded character we may without difficulty conclude that every function $f(X)$ which can be composed linearly of bounded characters of the group will have the same quality, *i.e.*, for every such function $f(X)$ the set of functions $\{f(X+H)\}$ will be compact. Now, guided by Bochner's formulation of the definition of the notion "almost periodicity" for the functions of a real variable, von Neumann set up the following general definition: A complex function $f(X)$ defined on an arbitrary Abelian group is called almost periodic on the group if the set of functions $\{f(X+H)\}$ is compact in the above sense.

As just mentioned, it is easily seen that every function on the group which can be composed linearly of bounded characters is almost periodic on the group. Von Neumann has shown that the converse theorem is valid for a quite arbitrary Abelian group (and not only for the translation group and the rotation group), so that the functions on an arbitrary Abelian group which can be linearly composed of the bounded characters of the group are exactly the almost periodic functions on the group. In its main ideas, von Neumann's proof of this fundamental theorem follows previous proofs given in the theory of the ordinary almost periodic functions. Corresponding to the theory of the ordinary almost periodic functions, where a Fourier series of the form $\sum A_n e^{i\Lambda_n x}$ was attached to every almost periodic function, there is also, in the general case of an arbitrary Abelian group, attached to any function $f(X)$ almost periodic on the group, a Fourier series, here of the form $\sum A_n \chi_n(X)$, where $\chi_1(X)$, $\chi_2(X)$, $\cdots$ is a denumerable set of bounded group characters characteristic of the function $f(X)$ under consideration. By a generalization of the method which Weyl developed to prove the uniqueness theorem for the ordinary almost periodic functions, it is shown that also in the general group theoretical case the Fourier series determines the function $f(X)$ uniquely. Thus, to two different almost periodic functions belong two different Fourier series. A generalization of the method applied by Wiener in proving the approximation theorem for the ordinary almost periodic functions is further used to accomplish the proof of the main theorem.

This theorem states that also in the general group theoretical case, starting from the Fourier series $\sum A_n\chi_n(X)$, we may form finite sums $s(X) = \sum a_n\chi_n(X)$ which approximate the given function $f(X)$ uniformly on the whole group.

So far, the theory of the almost periodic functions on arbitrary Abelian groups is quite parallel to the theory concerning the special case of the translation group. But before the theory could get started at all and be developed on the lines indicated above, there was a fundamental difficulty to be overcome which would seem to make the whole problem quite unapproachable. In fact the very basis of the formation of the Fourier series of an ordinary almost periodic function was the consideration of the mean value of the function $f(x)$ defined by

$$\lim_{\tau\to\infty} \frac{1}{2\tau}\int_{-\tau}^{\tau} f(x)dx.$$

In the general case, however, when we consider an arbitrary Abelian group without any topological structure, it might seem at first that there is no possibility of defining the notion of a mean value of a function over the group. I have no time to explain the simple and ingenious way in which von Neumann succeeded in defining this notion of the mean value $\mathfrak{M}\{f(X)\}$ of a function $f(X)$ almost periodic on the group. I shall only mention that, as soon as the mean value had been defined, the way was open to the further development of the theory in analogy with the theory of the ordinary almost periodic functions. In particular, as might be expected, the set of the bounded group characters proved to be a normalized orthogonal system, that is,

$$\mathfrak{M}\{\chi_1(X)\bar{\chi}_2(X)\} = \begin{cases} 0 & \text{for} \quad \chi_1 \neq \chi_2, \\ 1 & \text{for} \quad \chi_1 = \chi_2. \end{cases}$$

This being the case we naturally get the Fourier series $\sum A_n\chi_n(X)$ of an almost periodic function $f(X)$ by forming the mean value

$$a(\chi) = \mathfrak{M}\{f(X)\bar{\chi}(X)\}$$

for an arbitrary character $\chi(X)$ and by showing that this mean value is equal to 0 for all χ apart from a denumerable number of characters $\chi_n(X)$ characteristic for the function in question. These $\chi_n(X)$ are of course just the characters which occur in the composition of the almost periodic function $f(X)$ under consideration.

So far I have spoken as if the general theory of the almost periodic functions on an arbitrary Abelian group included in particular the ordinary theory of the purely periodic and of the almost periodic functions of a real variable, if we simply specialize our group to be the rotation group or the translation group, respectively. But as you may have observed, this is not the case. Indeed the pure oscillations forming the basis of the ordinary theory of the purely periodic or the almost periodic functions are not all of the bounded characters of the rotation group or the translation group, but only those bounded characters which are continuous on these particular groups with their natural topology. In the general case of an arbitrary Abelian group, however, we must necessarily treat all the bounded characters of the group on the same footing, as a restriction concerning continuity cannot be formulated at all on a non-topological group. Seen from a general group theoretical point of view, the different bounded characters are therefore all equally simple. In the special case of the translation group, the class of all von Neumann almost periodic functions on this group, which functions were investigated by Ursell prior to and independently of the general theory, is essentially more comprehensive than the class of the ordinary continuous almost periodic functions, as von Neumann does not claim continuity for his class. Fortunately, and this is another beautiful chapter of von Neumann's theory, the theory of the ordinary, *i.e.*, of the continuous almost periodic functions, proves to fit quite naturally into his general theory. I must confine myself to a few words about this point. Let us consider an arbitrary Abelian group, and let it be possible to introduce into this group (as it is possible in the case of the translation group and the rotation group) a topology of some kind or other in order to be able to ascribe any sense at all to the notion of continuous function on the group. Then the main theorem will remain valid if in the whole of the theorem we restrict the functions under consideration by demanding that they shall be continuous on the group considered as a topological group. More precisely, it holds for any topological group that the continuous functions $f(X)$ almost periodic on the group are just those functions which can be composed linearly of the continuous characters of the group.

5. Concluding remarks. In conclusion, I should like to make two remarks in order to emphasize what has been gained by von Neumann's general theory, of which I have given you only a rough outline. In the first place, and this must be said to be a characteristic feature of the mathematics of our day, we have achieved the combination and harmonization of investigations hitherto quite unrelated into one single theory of a general abstract character. Thus, in our case, we have learned about an intimate, hitherto unobserved, connection between the theory of Fourier series of purely periodic and almost periodic functions not only of one variable but of several variables, indeed of an infinite number of variables, and the theory of the group characters of the finite Abelian groups, in particular the group of the classes of residues, which forms the basis of the investigations of the distribution of the prime numbers in the different arithmetical progressions.

As for the other remark, it concerns in particular the translation group and the difference emphasized above between the theory of the almost periodic functions of this group considered as a group without any structure (apart of course from the structure given by the composition itself) and considered as a group topologized by means of the ordinary metric of the straight line. Von Neumann's theory has enabled us to fit into our considerations the total set of all the bounded characters of this group, *i.e.*, to operate not only with the continuous, but also with the discontinuous solutions of the classical functional equation

$$\phi(x + y) = \phi(x) \cdot \phi(y),$$

and to treat these discontinuous solutions, hitherto disdained, on exactly the same footing as the continuous solutions, *i.e.*, the pure oscillations. Indeed we have seen that in order to obtain the right systematization, it is even necessary to include these discontinuous solutions.

Here, as so often before in the history of mathematics, phenomena which appeared at first to be, so to speak, of a pathological nature, and which therefore from the start had to be excluded by means of protecting definitions, were later recognized, from a more general point of view, to be pertinent, even indispensable, to the subject under consideration.

ARE VARIABLES NECESSARY IN CALCULUS?*

KARL MENGER, Illinois Institute of Technology

1. The definite integral. We begin with a case in which the variables are superfluous beyond any doubt. By virtue of their definitions, the numbers

$$\int_0^\pi \sin x dx, \qquad \int_0^\pi \sin z dz, \qquad \int_0^\pi \sin \gamma d\gamma$$

are identical. Hence it does not make any difference which letter we use for the variable. But then why write the dummy part at all? We shall simply write

$$\int_0^\pi \sin.$$

A geometric consideration confirms this view. In a cartesian coördinate system, the sine function represents a curve, the sine curve

$$y = \sin x, \quad w = \sin z, \quad \text{or} \quad \delta = \sin \gamma,$$

according to whether the points are denoted by (x, y), (z, w) or (γ, δ). The numbers 0 and π determine an arc on the curve. The $\int$ sign indicates that we form the area under this arc. How we denote the points has no bearing on the area. We have

$$\int_0^\pi \sin = 2$$

---

* An address delivered before the Indiana Section at Purdue University, May 8, 1948. The author wishes to express his thanks to Mr. Burton D. Fried for the careful reading of the manuscript and numerous valuable suggestions.

and similarly

$$\int_{-1}^{1} \sinh = 0, \qquad \int_{1}^{2} \log = 2 \log 2 - 1.$$

If we wish to denote $\int_0^1 e^x dx$ without dummy variables we experience the first difficulty. For e^x denotes the value which the exponential function associates with the number x rather than the exponential function itself. But we remember that if in e^x we have to replace x by a long expression such as

$$-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma_1^2} - \frac{2\rho xy}{\sigma_1\sigma_2} + \frac{y^2}{\sigma_2^2}\right),$$

then typographical difficulties force us to print

$$\exp\left[-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma_1^2} - \frac{2\rho xy}{\sigma_1\sigma_2} + \frac{y^2}{\sigma_2^2}\right)\right]$$

where the symbol, exp, is used for the exponential function in the same way that the symbol, log, is used for the logarithmic function or the symbol, sin, for the sine function. In this notation we can write

$$\int_0^1 \exp = e - 1.$$

In eliminating the dummy part of $\int_0^1 x^n dx$ we are confronted with the complete lack of a symbol for the nth power function which associates the number x^n with x. If we feel that this important function deserves a symbol and we denote it by ${}^n\Pi$, then we can write

$$\int_0^1 {}^n\Pi = \frac{1}{n+1} \cdot$$

We do have a symbol for the less important nth root function, namely, $\sqrt[n]{}$. The typical polynomial has neither a name nor a symbol. The polynomial whose value for x is $a_0 + a_1 \cdot x + \cdots + a_n \cdot x^n$ might be denoted by $p_{a_0, a_1, \ldots, a_n}$. Polynomials of special importance may, of course, be denoted by special symbols. For instance, $p_{-1/2, 0, 3/2}$ is the second Legendre polynomial and is usually denoted by P_2. The frequent occurrence of the function associating with x the number $+\sqrt{1-k^2x^2}$ might warrant the introduction of a symbol. The fact that the graph of the function is the upper half of an ellipse of eccentricity $\sqrt{1-k^2}$ suggests the symbol $\text{ell}_{\sqrt{1-k^2}}$ for this function. In particular, $y = \sqrt{1-x^2}$ is the equation of a semi-circle, and we might write cir instead of ell_0. We should then have

$$\int_0^1 \text{cir} = \pi/4.$$

Instead of the constant polynomial p_c, which associates c with every x, we shall simply write c. For instance,

$$\int_0^\pi 2 = 2\cdot\pi, \qquad \int_0^\pi \sin 2 = \sin 2\cdot\pi.$$

All in all, what we need in the calculus of definite integrals are symbols for the most important functions rather than variables.

2. Substitution and the identity function. Instead of

$$\int_0^\pi \sin 2x dx = 0$$

we might write

$$\int_0^\pi \sin p_{0,2} = 0$$

where $\sin p_{0,2}$ denotes the function obtained by substituting the polynomial $p_{0,2}$ into the sine function. This notation for substitution is modelled after the symbol, log sin, in the classical theory where $\log \sin x$ denotes the value for x of the function obtained by substituting the sine into the logarithmic function. Denoting substitution by juxtaposition is unambiguous if we consistently use a dot in denoting a product. For instance, while $\sin p_{0,2}$ denotes the function whose value for x is $\sin 2x$ we write $\sin\cdot p_{0,2}$ for the function whose value for x is $\sin x\cdot 2x$. Similarly we distinguish between the functions log sin and log·sin assuming the values $\log \sin x$ and $\log x\cdot\sin x$, respectively.

But polynomials of the form $p_{0,c}$ are so frequently studied and the function $1/c\cdot p_{0,c}$, that is, the polynomial $p_{0,1} = {}^1\Pi$, is of such paramount importance that it seems to deserve a short symbol of its own. For $p_{0,1}$ is the function which associates x with x; the function which may be substituted into any function f without changing f; the function into which any function may be substituted without being changed. It is, in other words, the identity function, and the lack of a current symbol for this function strikingly illustrates how little heed we give to the algebraic aspects of calculus. We shall denote the identity function by j. The above integral reads

$$\int_0^\pi \sin(2\cdot j) = 0.$$

Denoting substitution by juxtaposition we can express the properties of j as follows

$$fj = jf = f \text{ for every } f\text{, in particular, } jj = j.$$

Incidentally, these equalities show that it would be incorrect to describe the introduction of j for the identity function as "just writing the letter j instead of the letter x." For could we in the classical notation write

$$fx = xf = f \quad \text{or} \quad f(x) = x(f) = f?$$

In particular, could we write

$$xx = x \quad \text{or} \quad x(x) = x?$$

As a matter of fact, the only way of transcribing the simple formula $fj=jf=f$ into the classical notation is by way of an implication of the following form: if $j(x)=x$ for every x, then $f(j(x))=j(f(x))=f(x)$ and, in particular, $j(j(x))=j(x)$.

The function j also enables us to define pairs of inverse functions, such as sin and arcsin, exp and log. We call g and g^* inverse functions if $gg^*=j$. (The functions g and g' are reciprocal if $g\cdot g'=1$.)

As long as we refrain from introducing a special symbol for the identity function, we are comparable to virtuosos in multiplication without a symbol for the number 1.

3. Differential calculus. It is obvious also that the formulae of differential calculus can be written without variables. If $\mathcal{D}f$ denotes the derivative of f, then the basic formulae read as follows:

$$\text{I.} \quad \mathcal{D}(f+g) = \mathcal{D}f + \mathcal{D}g;$$

$$\text{II.} \quad \mathcal{D}(f\cdot g) = f\cdot\mathcal{D}g + g\cdot\mathcal{D}f;$$

$$\text{III.} \quad \mathcal{D}(fg) = (\mathcal{D}f)g\cdot\mathcal{D}g.$$

In III, the term $\mathcal{D}(fg)$ denotes the derivative of the function obtained by substituting g into f, the term $(\mathcal{D}f)g$, the function obtained by substituting g into the derivative of f. By conventions about the scope of the symbol $\mathcal{D}$, we could dispense with parentheses in either one of the two expressions.

If we set

$$f = \text{logabs}, \quad \text{and} \quad g = \sin$$

where logabs is the function assuming the value $\log|x|$ for x, then we have

$$\mathcal{D}f = \text{rec} \quad \text{and} \quad \mathcal{D}g = \cos$$

where rec is the function ${}^{-1}\Pi$ associating with x the reciprocal number $1/x$, and we obtain from formula III

$$\mathcal{D}(\text{logabs}\sin) = \text{rec}\sin\cdot\cos = \cot.$$

4. The calculus of antiderivatives. We shall denote by $\mathcal{S}f$ any antiderivative of f, that is, any function having the derivative f. Hence we have

$$\textbf{A.} \quad \mathcal{D}(\mathcal{S}f) = f.$$

Moreover we shall write

B. $f \sim g$ if and only if $\mathcal{D}f = \mathcal{D}g$.

Obviously, the relation $\sim$ is reflexive, symmetrical, and transitive. We further readily prove

C. $\mathcal{S}f \sim g$ if and only if $f = \mathcal{D}g$,

D. $\mathcal{S}(\mathcal{D}f) \sim f$.

In this notation, the classical results concerning antiderivatives can be written without variables. For instance,

$$\mathcal{S}\ \cos \sim \sin, \qquad \mathcal{S}\ \exp \sim \exp, \qquad \mathcal{S}\ \text{rec} \sim \text{logabs}, \qquad \mathcal{S}\ \log \sim (j-1)\cdot\log.$$

5. Changing variables. The crucial test for a notation without variables is integration by substitution. For, traditionally, this method is treated as a change of variables. In our notation we have, first of all,

$$(\mathcal{S}\,h)g \sim \mathcal{S}\,[hg\cdot\mathcal{D}g]. \tag{1}$$

For, by **B** of Section 4, this formula is equivalent to

$$\mathcal{D}\{(\mathcal{S}\,h)g\} = \mathcal{D}\{\mathcal{S}\,[hg\cdot\mathcal{D}g]\}$$

and this last equality is true since both expressions are equal to $hg\cdot\mathcal{D}g$; the expression on the right side by virtue of **A** of Section 4; the expression on the left side since by virtue of Rules III of Section 3 and **A** of Section 4 we have

$$\mathcal{D}\{(\mathcal{S}\,h)g\} = [\mathcal{D}(\mathcal{S}\,h)]g\cdot\mathcal{D}g = hg\cdot\mathcal{D}g.$$

This completes the proof of (1). If on both sides of (1) we substitute an inverse of g, that is, a function g^* such $gg^*=j$, then, in view of $(\mathcal{S}\,h)j=\mathcal{S}\,h$, we obtain

$$\mathcal{S}\,h \sim [\mathcal{S}\,(hg\cdot\mathcal{D}g)]g^*$$

which is the formula for integration by substitution. The formula clearly indicates the four steps that we have to take in integrating h by the substitution of g, namely,

1) the substitution of g into h;
2) the multiplication by $\mathcal{D}g$;
3) the integration of the product;
4) the substitution of g^* into the antiderivative.

For instance, let h be rec cir, that is, the function associating the number $1/\sqrt{1-x^2}$ with x. If we wish to find an antiderivative of h by the substitution of the function $g=\sin$ for which $\mathcal{D}g=\cos$ and $g^*=\arcsin$, then, noting that cir sin $=\cos$ and $\mathcal{S}\,1 \sim j$, we obtain

$$\mathcal{S}\ \text{rec cir} \sim [\mathcal{S}\,(\text{rec}\cos\cdot\cos)]\arcsin \sim (\mathcal{S}\,1)\arcsin \sim j\arcsin = \arcsin.$$

Thus we do not need variables in order to "change variables."

It is important to realize that we can apply the method described above even if we refrain from introducing new symbols for special functions beyond j for the identity function. For instance, if we denote the function associating $1/\sqrt{1-x^2}$ with x by f and note that $1/\sqrt{1-\sin^2 t}=\sec t$ or, without variables, that $f\sin=\sec$, then we still have

$$\mathfrak{S}f \sim [\mathfrak{S}f\sin\cdot\mathcal{D}\sin]\arcsin \sim [\mathfrak{S}\sec\cdot\cos]\arcsin$$
$$\sim [\mathfrak{S}1]\arcsin \sim j\arcsin = \arcsin.$$

6. Functions of more variables. In what follows we shall confine ourselves to the study of functions of two variables and shall mainly stress a few points which are of importance for the calculus concerning functions of one variable. For more general considerations the reader is referred to other publications [1].

The relevance of functions of two variables for the theory of functions of one variable is illustrated by the following examples.

An implicitly defined function f of one variable is a function which, for some function F of two variables, satisfies the condition $F(x, f(x))=0$ for every x. Functions of one variable which are solutions of a first order differential equation present similar problems of notation.

Some important functions of two variables are defined as integrals with regard to a parameter, that is, of functions of two variables with regard to one of them. For instance, the value of the Gamma function for x is defined as

$$\Gamma(x) = \int_0^\infty e^{-y}\cdot y^{x-1}dy.$$

Laplace transforms and integral equations lead to similar expressions.

The difference quotients of a function f of one variable are the values of a function of two variables, *viz.*, the function

$$\frac{f(y)-f(x)}{y-x}.$$

Similarly, the definite integral $\int_x^y f(t)dt$ of a function of one variable between arbitrary limits x and y is a function of two variables which, without dummy variables, can be written in the form $\int_x^y f$.

Functional equalities for functions of one variable are connected with functions of more variables. For instance, the equality for the exponential function,

$$e^x\cdot e^y = e^{x+y} \quad\text{or}\quad \exp x\cdot\exp y = \exp(x+y),$$

is related to a function of two variables. The function f is homogeneous of degree n if

$$f(x\cdot y) = f(x)\cdot y^n.$$

Finally, even the basic operations of adding and multiplying two functions, f and g, can be interpreted as the result of substituting f and g into a sum-function S and a product-function P, respectively. Here S and P are functions of two variables for which

$$S(x, y) = x + y, \qquad P(x, y) = x \cdot y.$$

7. Substitution in the realm of functions of two variables. Besides the functions f, g, sin, exp, $\cdots$ of one variable we shall now also admit functions of two variables which we shall denote by capital letters $F, G, S, P, \cdots$. With regard to the substitution of i functions of k variables into a function of i variables we shall make the following conventions (ik):

(11). By substituting g into f we obtain a function fg of one variable;

(12). By substituting G into f we obtain a function fG of two variables; if $f=\sin$ and $G=S$, then $fG=\sin S$ is the function associating the number $\sin(x+y)$ with (x, y);

(21). By substituting (g, h) into F we obtain a function $F(g, h)$ of one variable; if $F=P$, $g=\sin$, $h=\exp$, then $F(g, h)=P(\sin, \exp)$ is the function associating the number $\sin t \cdot \exp t$ with t.

(22). By substituting (G, H) into F we obtain a function $F(G, H)$ of two variables. If $F=P$, $G=S$, $H=\sin P$, then $F(G, H)=P(S, \sin P)$ is the function associating the number $(x+y)\cdot\sin(x\cdot y)$ with (x, y).

Each of these substitutions satisfies important one-side distributive laws in conjunction with addition and multiplication, namely,

$$(f_1 + f_2)\gamma = f_1\gamma + f_2\gamma \quad \text{and} \quad (f_1 \cdot f_2)\gamma = f_1\gamma \cdot f_2\gamma$$

where γ may be either a function g of one variable or a function G of two variables, and

$$(F_1 + F_2)(\gamma, \delta) = F_1(\gamma, \delta) + F_2(\gamma, \delta) \quad \text{and} \quad (F_1 \cdot F_2)(\gamma, \delta) = F_1(\gamma, \delta) \cdot F_2(\gamma, \delta)$$

where (γ, δ) may be either a pair (g, h) or a pair (G, H).

Interpreting F as a surface $z=F(x, y)$ in the same way as we have interpreted f as a curve $y=f(x)$ we see that

$$z = \sin x \cdot \exp x \quad \text{and} \quad z = \sin y \cdot \exp y$$

are different surfaces. Both surfaces are cylindrical, their cross-sections are congruent, but their generating lines are perpendicular. Does the variable matter, after all?

The explanation lies in the generalization of the identity function j of one variable, which associates x with x. In the realm of functions of two variables, there are two functions, J_1 and J_2, such that

$$J_1(x, y) = x \quad \text{and} \quad J_2(x, y) = y \qquad \text{for every } (x, y).$$

We have

$$F(J_1, J_2) = F, \qquad J_1(\gamma, \delta) = \gamma, \qquad J_2(\gamma, \delta) = \delta$$

for every function F and every pair (g, h) and (G, H). If a function F of two variables has the property that $F(j, h) = F$ for every function h, then, in the classical terminology, F is a function of x and y depending on x only. The condition $F = F(j, 0)$ is sufficient for F to have the above property. An example of a function of this kind is J_1 and, in fact, fJ_1 for every function f of one variable. Similarly, $J_2 = J_2(0, j)$ is an example of a function of two variables depending upon the second variable only.

The two different cylindrical surfaces mentioned before correspond to two different functions of two variables, namely,

$$(\sin \cdot \exp)J_1 = \sin J_1 \cdot \exp J_1$$

and

$$(\sin \cdot \exp)J_2 = \sin J_2 \cdot \exp J_2,$$

respectively. After what was said in Section 3 about j it should not be necessary to emphasize that the notation without variables for the two functions does not amount to just replacing the letters x and y by the letters J_1 and J_2. We also see that

$$S = J_1 + J_2 \quad \text{and} \quad P = J_1 \cdot J_2.$$

The difference quotients of a function f of one variable now appear to be the values of the following function of two variables

$$\frac{fJ_2 - fJ_1}{J_2 - J_1}.$$

The functional equality of the exponential function reads

$$\exp J_1 \cdot \exp J_2 = \exp (J_1 + J_2) \quad \text{or} \quad P(\exp J_1, \exp J_2) = \exp S.$$

The function f is homogeneous of degree n if, and only if,

$$f(J_1 \cdot J_2) = fJ_1 \cdot {}^n\Pi J_2 \quad \text{or} \quad fP = P(fJ_1, {}^n\Pi J_2).$$

As examples of the role of j in the realm of functions of two variables, we mention the implicit definition of a function f of one variable by the equality $F(j, f) = 0$ for some function F and the first order differential equation $\mathcal{D}f = F(j, f)$ for a given function F.

The differential equation which is traditionally written in the form

$$(\alpha) \qquad M(x, y)dx + N(x, y)dy = 0$$

where M and N are given functions, can be interpreted in one of the following three ways:

(1) as the condition $M(j, y) + N(j, y) \cdot \mathcal{D}y = 0$ for a function y of one variable;

(2) as the condition $M(x, j) \cdot \mathcal{D}x + N(x, j) = 0$ for a function x of one variable;

(3) as the condition

$$M(x, y)\cdot\mathcal{D}x + N(x, y)\cdot\mathcal{D}y = 0 \tag{β}$$

for a pair of functions x, y of one variable which are traditionally written in the form $x(t)$ and $y(t)$. We have here used the letters x and y rather than f and g as symbols for functions, in order to avoid the impression that we attribute to the first letters of the alphabet properties not shared by the last letters. Yet, the resemblance of (β) with the classical form (α) is somewhat illusive. For in (β), x and y do not mean variables, and $\mathcal{D}x$ and $\mathcal{D}y$ do not mean differentials.

8. Partial derivatives. There are two operators of differentiation in the realm of functions of two variables, $\mathcal{D}_1$ and $\mathcal{D}_2$. In the classical theory, $\mathcal{D}_1F$ and $\mathcal{D}_2F$ are called the two partial derivatives of F with regard to the first and the second variable, respectively. As before, the derivative of the function f of one variable will be denoted by Df. In particular, we have

$$\mathcal{D}_1J_1 = \mathcal{D}_2J_2 = 1 \quad \text{and} \quad \mathcal{D}_2J_1 = \mathcal{D}_1J_2 = 0. \tag{2}$$

Now let $\int f$ be the function of two variables associating with (x, y) the definite integral $\int_x^y f$. (There is no danger of confusing $\int f$ with an antiderivative of f since the latter function has been denoted by $\mathfrak{S}f$.) The definite integral $\int_0^\pi \sin$, studied in Section 1, is the value of the function $\int \sin$ for $(0, \pi)$. Hence, in a systematic notation we should write

$$\left(\int \sin\right)(0, \pi) = 0, \qquad \left(\int \log\right)(1, 2) = 2\cdot\log 2 - 1, \text{ etc.}$$

The computation of a definite integral by substitution, written without variables, reads

$$\int f = \left(\int fg\cdot\mathcal{D}g\right)(g^*J_2, g^*J_1).$$

The fundamental laws concerning the reciprocity of differentiation and definite integration, traditionally written in the form

$$\int_a^b f'(t)dt = f(b) - f(a) \quad \text{or} \quad \int_a^b f(t)dt = g(b) - g(a) \quad \text{where} \quad g'(t) = f(t)$$

and

$$\frac{d}{dy}\int_a^y f(t)dt = f(y) \quad \text{and} \quad \frac{d}{dx}\int_x^b f(t)dt = -f(x),$$

can now be written without variables

$$\int (\mathcal{D}f) = fJ_2 - fJ_1 \quad \text{or} \quad \int \mathcal{D}f = (\mathfrak{S}f)J_2 - (\mathfrak{S}f)J_1$$

where $\mathcal{S}f$ is any antiderivative of f, but the same in both places, and

$$\mathcal{D}_1 \int f = -fJ_1, \qquad \mathcal{D}_2 \int f = fJ_2.$$

Partial differentiation is connected with addition and multiplication by the analogues of formulae I and II of Section 3,

I_i $\mathcal{D}_i(F+G) = \mathcal{D}_iF + \mathcal{D}_iG$,

II_i $\mathcal{D}_i(F\cdot G) = F\cdot\mathcal{D}_iG + G\cdot\mathcal{D}_iF$.

Differentiation is connected with substitution by the following generalizations of Formula III of Section 3.

III_{12}. $\mathcal{D}_i(fG) = (\mathcal{D}f)G\cdot\mathcal{D}_iG \qquad (i = 1, 2).$

Example. If $G = P = J_1\cdot J_2$, then, in view of formulae (2),

$$\mathcal{D}_2(fP) = J_1\cdot(\mathcal{D}f)P.$$

If f is homogeneous of degree n (*cf.* Section 6), then

$$fP = fJ_1\cdot {}^{n}\Pi J_2,$$

and thus

$$\mathcal{D}_2(fP) = fJ_1\cdot n\cdot {}^{n-1}\Pi J_2.$$

Hence

$$fJ_1\cdot n\cdot {}^{n-1}\Pi J_2 = J_1\cdot(\mathcal{D}f)P$$

and, noting that

$$J_1(j, 1) = j, \qquad J_2(j, 1) = 1, \qquad P(j, 1) = j, \qquad fj = f, \qquad (\mathcal{D}f)j = \mathcal{D}f,$$

we obtain Euler's relation

$$n\cdot f = j\cdot\mathcal{D}f.$$

III_{21}. $\mathcal{D}[F(g, h)] = (\mathcal{D}_1F)(g, h)\cdot\mathcal{D}g + (\mathcal{D}_2F)(g, h)\cdot\mathcal{D}h.$

Example. If $g = j$ and thus $\mathcal{D}g = \mathcal{D}j = 1$, then

$$\mathcal{D}[F(j, h)] = (\mathcal{D}_1F)(j, h) + (\mathcal{D}_2F)(j, h)\cdot\mathcal{D}h.$$

If h is implicitly defined by $F(j, h) = 0$, then $\mathcal{D}[F(j, h)] = \mathcal{D}0 = 0$ and we obtain

$$\mathcal{D}h = -(\mathcal{D}_1F)(j, h)/(\mathcal{D}_2F)(j, h)$$

for the derivative of the implicit function h.

III_{22}. $\mathcal{D}_i[F(G, H)] = (\mathcal{D}_1F)(G, H)\cdot\mathcal{D}_iG + (\mathcal{D}_2F)(G, H)\cdot\mathcal{D}_iH \qquad (i = 1, 2).$

Example. If we define a function H implicitly by $F(x, H(x, y)) = y$, then, without variables $F(J_1, H) = J_2$. Using formulae (2) from III_{22}, we obtain the expressions for $\mathcal{D}_1 H$ and $\mathcal{D}_2 H$.

As has been shown, *l.c.* [1], Formula III and its extensions to functions of any number of variables can be synthesized.

9. Partial integration. Corresponding to the two operators $\mathcal{D}_1$ and $\mathcal{D}_2$, there are two operators $\mathcal{S}_1$ and $\mathcal{S}_2$ of antiderivation and two partial definite integrals $\int_1 F$ and $\int_2 F$. For instance, the Laplace transform of a function f

$$\int_0^\infty \exp(-x\cdot y)\cdot f(y)dy$$

is a definite integral with regard to the second variable of a function of two variables which, without variables, can be written in the form

$$\exp(-J_1\cdot J_2)\cdot fJ_2 \quad \text{or} \quad \exp(-P)\cdot fJ_2.$$

The integral is a function of two variables depending only upon the first. If, in this function, we substitute $(j, 0)$ and observe that $gj = g$ we obtain a function of one variable which we may denote by

$$\mathcal{L}f = \left[\int_{0^2}^\infty \exp(-P)\cdot fJ_2\right](j, 0).$$

Integral equations and the Gamma function can be written in a similar way.

10. Conclusions. While variables are not necessary for the presentation of fundamental results of calculus, there remain two questions. To what extent are variables necessary in proving these results? And, are variables not desirable even in formulating the theorems?

Since most students learn calculus as a tool, and since books on physics, engineering, statistics, mathematical economics, etc., are written in the classical notation, it is clear that, in initiating students into calculus, we have to use the classical notation. Yet I feel that the possibility of a consistent notation without variables should influence our teaching, namely, in the direction of reducing the use of variables. I further think that, at least in a few cases, we should mention the alternative form and, in particular, make the student aware of the possibility of a consistent notation which dispenses with dummy variables. I even believe that the ability to grasp, say, integration by substitution without variables (Sections 5 and 8) is a gauge for a student's real understanding of calculus.

In proving formulae, we shall make use of variables although perhaps again at a diminishing rate. In proving, for instance, Formula III of Section 3 we show that if for a number x_0 the three numbers

$$\mathcal{D}g, \qquad (\mathcal{D}f)g, \qquad \mathcal{D}(fg)$$

are meaningful, then the third is the product of the first two. (In fact, we prove even more.) This result may be interpreted in the following form: At a place where the three terms of formula III are meaningful, the formula is true. Many formulae can be interpreted in the sense that they are true provided that every term is meaningful. For elementary functions, one could even develop an algebra of their domains of definition accompanying the algebras of their substitution and differentiation.

Another point brought out by these developments is that the application of the limit concept can be confined to the proof of very few basic formulae from which all the other formulae can be obtained by some algebra.

In concluding we mention the existence of finite models for the algebra of analysis [2] comparable to finite rings or fields enjoying the essential properties of the system of all real numbers, and to finite planes which have the essential properties of our affine or projective plane.

References

1. The author's *Algebra of Analysis*, Notre Dame Mathematical Lectures, No. 3, 1944.

2. *Cf.* the author's *Tri-Operational Algebra*, Reports of a Mathematical Colloquium, Issue 5–6, Notre Dame, 1944; *General Algebra of Analysis*, Issue 7, 1946; and numerous other papers in the same issues.

A GENERALIZATION OF FEUERBACH'S THEOREM

H. F. SANDHAM, Dublin, Ireland

1. Introduction. The theorem that a circle touching the sides of a triangle touches the nine-point circle is due to Feuerbach. A generalization due to Hart which seems to complete the theorem in one direction is that four circles touching four given circles touch a fifth. In the geometry of the triangle many generalizations have been given, but one which is not a statement about conics is McCay's [1], that the pedal circle of two isogonal conjugates which are collinear with the circumcenter touches the nine-point circle.

The theorem of this paper is a generalization of the last, and may be stated in two ways:

1. *The angle between the nine-point circle of a triangle and the pedal circle of two isogonal conjugates is equal to the angle which the line joining the points subtends at the inverse of either in the circumcircle.*

2. *The angle between the nine-point circle of a triangle ABC and the pedal circle of a point P is equal to*

$$\measuredangle PBC + \measuredangle PCA + \measuredangle PAB \pm \pi/2.$$

The proof is a deduction from three formulas found in *Inversive Geometry* by

Morley and Morley.

2. Angle between two circles. Since the proof is by complex numbers it is convenient to regard the angle between two lines, denoted by $\sphericalangle AOB$, as a measure of the anticlockwise rotation which brings OB into coincidence with OA [2], and thus to have the convention

$$\sphericalangle AOB = - \sphericalangle BOA.$$

If two intersecting circles have radii ρ_1, ρ_2 and the distance between their centers is δ_{12}, and if θ is the angle, positive or negative, subtended by the line joining the centers at either point of intersection, then

$$2\rho_1\rho_2 \cos\theta = \rho_1^2 + \rho_2^2 - \delta_{12}^2.$$

It is convenient to take this as the definition of the angle between two circles [2], and thus to make no distinction between $\pm\theta\pm 2n\pi$.

3. Angle between pedal circle and nine-point circle. Taking the circumcenter as origin and the circumradius as unity, if the vertices of a triangle are denoted by the complex numbers t_1, t_2, t_3, then

$$t_1\bar{t}_1 = t_2\bar{t}_2 = t_3\bar{t}_3 = 1.$$

When the Euler line is taken as the axis of real numbers

$$t_1 + t_2 + t_3 = \bar{t}_1 + \bar{t}_2 + \bar{t}_3.$$

If x, y are isogonal conjugates and ρ is the radius of their pedal circle, the three formulas [3] mentioned are

$$x + y + \bar{x}\bar{y}s_3 = s_1 \tag{1}$$

$$\bar{x} + \bar{y} + xy/s_3 = s_1 \tag{2}$$

$$(1 - x\bar{y})(1 - \bar{x}y) = 4\rho^2 \tag{3}$$

where

$$s_1 = t_1 + t_2 + t_3, \qquad s_2 = t_2t_3 + t_3t_1 + t_1t_2, \qquad s_3 = t_1t_2t_3.$$

Since $(x+y)/2$ is the center of the pedal circle and $s_1/2$ is the center of the nine-point circle, and its radius is $1/2$, the angle between the pedal circle and the nine-point circle is given by

$$\tfrac{1}{2}\left|1 - x\bar{y}\right|\cos\theta = \tfrac{1}{4} + \tfrac{1}{4}(1 - x\bar{y})(1 - \bar{x}y) - \tfrac{1}{4}(x + y - s_1)(\bar{x} + \bar{y} - s_1).$$

Substitution from (1) and (2) gives

$$\begin{aligned}\tfrac{1}{2}\left|1 - x\bar{y}\right|\cos\theta &= \tfrac{1}{4} + \tfrac{1}{4}(1 - x\bar{y})(1 - \bar{x}y) - \tfrac{1}{4}xy\bar{x}\bar{y}\\ &= \tfrac{1}{4}(1 - x\bar{y} + 1 - \bar{x}y).\end{aligned}$$

Thus θ is equal to the amplitude of $1 - x\bar{y}$. This is the basis of the discussion.

4. First statement of theorem. Since the equation of the circumcircle is

$$t\bar{t} = 1,$$

then, if y' denote the inverse of y,

$$y'\bar{y} = 1.$$

Hence the angle between the nine-point circle and the pedal circle of x, y has been proved equal to

$$\operatorname{amp}(y' - x)/y' = \operatorname{amp}(y' - x)/(y' - y).$$

This is the first statement of the theorem.

5. Second statement of theorem. Elimination of y from (1) and (2) gives

$$\bar{y} = (x^2/s_3 - xs_1/s_3 + s_1 - \bar{x})/(1 - x\bar{x}),$$

so that

$$\begin{aligned} 1 - x\bar{y} &= (s_3 - s_2x + s_1x^2 - x^3)/s_3(1 - x\bar{x}) \\ &= (t_1 - x)(t_2 - x)(t_3 - x)/t_1t_2t_3(1 - x\bar{x}). \end{aligned}$$

The amplitude of

$$(t_1 - x)(t_2 - x)(t_3 - x)/t_1t_2t_3$$

is the (algebraic) sum of the angles the line joining x and the circumcenter subtends at the vertices, and is equal to

$$\measuredangle\, xt_2t_3 + \measuredangle\, xt_3t_1 + \measuredangle\, xt_1t_2 - \pi/2.$$

The expression

$$1 - x\bar{x}$$

is of course real, and is negative if x is outside the circumcircle, and positive if x is inside. Thus

$$\theta = \measuredangle\, xt_2t_3 + \measuredangle\, xt_3t_1 + \measuredangle\, xt_1t_2 \pm \pi/2,$$

where the sign is plus if x is outside the circumcircle and minus if x is inside.

This is the second statement of the theorem.

References

1. W. S. McCay, Transactions of the Royal Irish Academy, vol. xxix.
2. R. A. Johnson, Modern Geometry, Houghton Mifflin (1929), pp. 11, 128.
3. F. Morley and F. V. Morley, Inversive Geometry, Ginn (1933), pp. 196, 197.

MATHEMATICAL NOTES

EDITED BY E. F. BECKENBACH, University of California, Los Angeles

Materials for this department should be sent direct to E. F. Beckenbach, University of California, Los Angeles 24, California.

ON THE CONVERSE OF FERMAT'S THEOREM

P. ERDÖS, University of Illinois

Following Lehmer we shall call an integer n a *pseudoprime* if $2^n \equiv 2 \pmod{n}$ and n is not a prime. The smallest pseudoprime is $341 = 11.31$. Recently Sierpinski[1] gave a very simple proof that there are infinitely many pseudoprimes, by proving that if n is a pseudoprime then $2^n - 1$ is also a pseudoprime. Lehmer[2] proved that there exist infinitely many pseudoprimes n with $v(n) = 3$, where $v(n)$ denotes the number of different prime factors of n. In the present note we prove the following theorem.

THEOREM. *For every k there exist infinitely many squarefree pseudoprimes with $v(n) = k$.*

First we repeat Lehmer's proof[3] that there are infinitely many pseudoprimes n with $v(n) = 2$. It is well known[4] that for every $m > 6$ both $2^m - 1$ and $2^m + 1$ have a primitive prime factor; that is, there exist primes p and q such that

$$2^m - 1 \equiv 0 \pmod{p}, \qquad 2^l - 1 \not\equiv 0 \pmod{p}, \qquad \text{for } 1 \leqq l < m;$$
$$2^m + 1 \equiv 0 \pmod{q}, \qquad 2^l + 1 \not\equiv 0 \pmod{q} \qquad \text{for } 1 \leqq l < m.$$

It is easy to see that $p \cdot q$ is a pseudoprime. In fact we have $p \equiv q \equiv 1 \pmod{2m}$, $2^{2m} \equiv 1 \pmod{p \cdot q}$, thus

$$2^{pq-1} \equiv 2^{(p-1)(q-1)} \cdot 2^{p-1} \cdot 2^{q-1} \equiv 1 \pmod{pq}.$$

Also it is immediate that to different values of m correspond different values of $p \cdot q$, which proves the theorem for $k = 2$.

The proof of the general case will be very similar to that of Lehmer. We use induction on k. Let $n_1 < n_2 < \cdots$ be an infinite sequence of pseudoprimes with $v(n_i) = k - 1$. Let p_i be one of the primitive prime factors of $2^{n_i - 1} - 1$. We claim that $p_i \cdot n_i$ is a pseudoprime. In fact, by definition, $2^{n_i - 1} \equiv 1 \pmod{p_i \cdot n_i}$, also $2^{p_i - 1} \equiv 1 \pmod{p_i}$. Further, since $p_i - 1 \equiv 0 \pmod{(n_i - 1)}$, we have $2^{p_i - 1} \equiv 1 \pmod{n_i}$ and finally $2^{n_i - 1} \equiv 1 \pmod{n_i}$. Thus

[1] Colloquium Math., vol. 1 (1947), p. 9.

[2] This MONTHLY, vol. 56 (1949) p. 306.

[3] This MONTHLY, vol. 43 (1936), pp. 347–356.

[4] Bang, Tiddsskrift for Mat. 1886, pp. 130–137. See Also Birkhoff-Vandiver, Annals of Math., 1904.

$$2^{n_i p_i - 1} = 2^{(n_i-1)(p_i-1)} \cdot 2^{n_i-1} \cdot 2^{p_i-1} \equiv 1 \pmod{p_i n_i}.$$

Also $p_i > n_i$, since $p_i \equiv 1 \pmod{(n_i - 1)}$ and n_i is not a prime. Thus $p_i \cdot n_i$ is squarefree, and $v(p_i \cdot n_i) = k$, and all the integers $p_i \cdot n_i$ are different; this completes the proof of the theorem.

Following Lehmer we call n an absolute pseudoprime if $a^n \equiv a \pmod{n}$ for every a prime to n. The smallest absolute pseudoprime is 561. It seems very difficult to determine whether there are infinitely many absolute pseudoprimes. A similar question is whether there exist any composite numbers n with $n-1 \equiv 0 \pmod{\phi(n)}$.

Two further questions are: Are there integers n so that $2^n - 1$ has more than k primitive prime factors? Are there infinitely many primes p for which $2^p - 1$ is composite? The smallest such prime is 11.

Denote by $f(x)$ the number of pseudoprimes not exceeding c. I can prove that

$$c_1 \cdot \log x < f(x) < c_2 \frac{x}{(\log x)^k}, \tag{1}$$

for every k if x is sufficiently large. In other words, the number of pseudoprimes is considerably smaller than the number of primes. The proof of (1) (second inequality) is complicated and we do not discuss it here.

A THEOREM ON THE DISTRIBUTION OF PRIMES

LEO MOSER, Universities of Manitoba and North Carolina

Using elementary properties of integers only, we shall establish the following result.

THEOREM: *For every positive integer r, there exists a prime p, with $3 \cdot 2^{2r-1} < p < 3 \cdot 2^{2r}$.*

This theorem is almost as strong as Bertrand's postulate, which states that for every real value of $x \geqq 1$ there is a prime p with $x < p \leqq 2x$.

The following four lemmas follow easily from Legendre's expression for $n!$, namely,

$$n! = \prod_p p^{\sum_{i=1}^{\infty} [n/p^i]}$$

They are proved in [1], which contains an exposition of Erdös' proof of Bertrand's postulate.

(1) *If $n < p \leqq 2n$, then p occurs exactly once in $\binom{2n}{n}$.*

(2) *If $n \geqq 3$ and $2n/3 < p < n$, then p does not occur in $\binom{2n}{n}$.*

(3) *If* $p^2 > 2n$, *then* p *occurs at most once in* $\binom{2n}{n}$.

(4) *If* $2^a \leqq 2n < 2^{a+1}$, *then* p *occurs at most* a *times in* $\binom{2n}{n}$.

Suppose there is no prime p with

(5) $$3.2^{2r-1} < p < 3.2^{2r};$$

then

(6) $$\binom{3.2^{2r}}{3.2^{2r-1}} \leqq \binom{2^{2r}}{2^{2r-1}}\binom{2^{2r-1}}{2^{2r-2}}\cdots\binom{2}{1}\left\{\binom{2^{r+1}}{2^r}\binom{2^r}{2^{r-1}}\cdots\binom{2}{1}\right\}^{2r}.$$

By (1), (2), (3), and (5), every prime which appears on the left-hand side of (6) appears also on the right; and those primes which appear with multiplicity greater than 1 on the left appear on the right with multiplicity at least $2r+1$, which by (4) is at least equal to the multiplicity with which they appear on the left.

On the other hand, a simple computation shows that for $r \geqq 6$, we have

(7) $$3.2^{2r} > (2^{2r} + 2^{2r-1} + \cdots + 2) + 2r(2^{r+1} + 2^r + \cdots + 2).$$

If we now interpret $\binom{2n}{n}$ as the number of ways of choosing n objects from $2n$, it follows from (7) that the inequality in (6) should be reversed. Hence for $r \geqq 6$ we have a contradiction which proves the theorem for these values of r. The primes 7, 29, 97, 389, and 1543 show that the theorem is also true for $r < 6$.

Our method may also be used to prove Bertrand's Postulate. For this purpose, if $2^\alpha < 2n \leqq 2^{\alpha+1}$, we replace (6) by

$$\binom{2n}{n} \leqq \binom{2a_1}{a_1}\binom{2a_2}{a_2}\cdots\binom{2}{1}\left\{\binom{2a_k}{a_k}\binom{2a_{k+1}}{a_{k+1}}\cdots\binom{2}{1}\right\}^{\alpha},$$

where

$$a_1 = \left[\frac{n+1}{3}\right], \qquad a_i = \left[\frac{a_{i-1}+1}{2}\right] \qquad \text{for } i > 1, \text{ and } k = \left[\frac{\alpha+1}{2}\right].$$

The details of the proof in this case can be filled in by the reader. Although the resulting proof is not quite as neat as the proof of the somewhat weaker result, it is still, we believe, simpler than the known proofs of Bertrand's Postulate [1, 2, 3].

References

1. R. G. Archibald, Bertrand's Postulate, Scripta Mathematica, vol. 11 (1941) pp. 109–120.

2. E. Landau, Handbuch der Lehre von der Verteilung der Primzahlen, vol. 1, Leipzig and Berlin, 1909, pp. 89–92.

3. S. Ramanujan. Collected Papers, pp. 208–209.

FERMAT'S EQUATION AND TSHEBYSHEFF'S POLYNOMIALS

ALAN WAYNE, Flushing, N. Y.

There is an interesting relation between Fermat's equation and the Tshebysheff polynomials which seems, so far, to have gone unstated.

It will be recalled that Fermat's equation (also known, inappropriately, as Pell's equation) is

$$v^2 - aw^2 = 1, \tag{1}$$

where v and w are integers, and a is a positive constant which is not the square of a rational number. The two smallest positive integers v_1 and w_1 which satisfy (1) constitute its *fundamental solution*. It is known that such a fundamental solution always exists, for if $\sqrt{a}$ be expanded into a simple continued fraction, there exist positive convergents v_m/w_m such that v_m and w_m satisfy (1), and v_1/w_1 is the least of these convergents. (See pages 36–38 of Harry N. Wright, *First Course in the Theory of Numbers*, John Wiley and Sons, New York, 1939.)

It is known also that all solutions v_n and w_n of (1), in zero or integers, without any exception, are obtained by equating the rational and irrational parts in the relation

$$v_n + w_n\sqrt{a} = \pm (v_1 + w_1\sqrt{a})^n, \tag{2}$$

where n is any integer or zero. (See page 351 of Uspensky and Heaslet, *Elementary Number Theory*, McGraw-Hill, New York, 1939.)

Now let us put $v_1 = \cos\theta$. Then, from (1), $w_1 = -i \sin\theta/\sqrt{a}$. Substituting these results in (2), and making use of De Moivre's Theorem, we find that $v_n + w_n\sqrt{a} = \pm(\cos n\theta - i \sin n\theta)$.

Let $v_n = \pm\cos n\theta$. Then $w_n = \mp i \sin n\theta/\sqrt{a}$, and again making use of (1), we have $w_n = (\pm \sin n\theta/\sin\theta)w_i$. Thus $v_n = \pm T_n(v_1)$, and $w_n = \mp w_1 U_{n-1}(v_1)$, where $T_n(x)$ and $U_n(x)$ are the Tshebysheff polynomials of the first kind and second kind, respectively, defined by $x = \cos\theta$, $T_n(x) = \cos n\theta$, and $U_n(x) = (\sin (n+1)\theta)/\sin\theta$, for integer or zero n. (See pages 55–56 of Madelung and Erwin, *Die Mathematischen Hilfsmittel des Physikers*, Dover Publications, New York, 1943.)

A NOTE ON FERMAT'S CONGRUENCE

V. L. KLEE, JR.,* University of Virginia

Carmichael [1, p. 53] defines a function λ as follows (ϕ being Euler's function): $\lambda(2^e) = \phi(2^e)$ if $0 \leqq e \leqq 2$; $\lambda(2^e) = \frac{1}{2}\phi(2^e)$ if $e \geqq 3$; $\lambda(p^e) = \phi(p^e)$ if p is an odd prime; $\lambda(2^e p_1^{e_1} \cdots p_n^{e_n})$ is the least common multiple of $\lambda(2^e)$, $\lambda(p_1^{e_1}), \cdots,$ $\lambda(p_n^{e_n})$, where $2, p_1, \cdots, p_n$ are distinct primes. He proves (p. 54) that if a is prime to m, then $a^{\lambda(m)} \equiv 1 \pmod{m}$; and later (pp. 72–74) he shows that each integer m has a primitive λ-root—that is, an integer which belongs modulo m to

* I am indebted to the referee for helpful suggestions regarding the arrangement of the proof.

the exponent $\lambda(m)$. Now denote by $L(m)$ the set of all integers k such that $a^k \equiv 1 \pmod{m}$ whenever a is prime to m. Then from Carmichael's results it follows that $L(m)$ consists of all multiples of $\lambda(m)$.

Now let us, in contrast, fix our attention on a definite exponent n, and consider the congruence, $a^n \equiv 1 \pmod{h}$, for various values of h. Denote by $C(n)$ the set of all integers h such that this congruence holds whenever a is prime to h, and let $\gamma(n)$ denote the greatest integer in $C(n)$. (We shall show that $C(n)$ is bounded.) As will be indicated below, γ is closely related to another function considered by Carmichael [**2**].

In this note we establish some "dual" relations between L and λ on the one hand and C and γ on the other. The first such relation, which is an immediate consequence of the definitions, is:

(1) $n \in L(m)$ if and only if $m \in C(n)$.

We next establish the following result:

(2) $L(m)$ consists of all multiples of $\lambda(m)$, $C(n)$ of all divisors of $\lambda(n)$.

The proof of (2) for L and λ was indicated above. To prove the statement for C and γ we note first that $\phi(x) \to \infty$ as $x \to \infty$, whence $\lambda(x) \to \infty$ as $x \to \infty$. Now let $\gamma^*(m)$ denote the greatest integer x for which $\lambda(x) \mid n$. (By the preceding remark, $\gamma^*(n)$ is defined for each n.) Consider an arbitrary $h \in C(n)$ and let r be a primitive λ-root of h. From the congruence, $r^n \equiv 1 \pmod{h}$, we have $\lambda(h) \mid n$, whence $h \leqq \gamma^*(n)$ and $C(n)$ is bounded. Now if $d \mid \gamma^*(n)$, we have $\lambda(d) \mid \lambda(\gamma^*(n))$, whence $\lambda(d) \mid n$ and $d \in C(n)$. Hence $\gamma(n) = \gamma^*(n)$ and $C(n)$ contains all divisors of $\gamma(n)$.

It remains to show that $C(n)$ contains only divisors of $\gamma(n)$. Suppose that $h \in C(n)$, $h \nmid \gamma(n)$, and let b denote the least common multiple of h and $\gamma(n)$. Since $b > \gamma(n)$, there is an integer a prime to b such that $a^n \not\equiv 1 \pmod{b}$. But we also have $a^n \equiv 1 \pmod{h}$ and $a^n \equiv 1 \pmod{\gamma(n)}$, whence $a^n \equiv 1 \pmod{b}$, which is a contradiction. Thus the proof of (2) is complete.

Immediately from (1) and (2) we have the following results.

(3) $m \mid \gamma(n)$ if and only if $\lambda(m) \mid n$.

(4) $L(m)$ may also be characterized as the set of all integers k for which $m \mid \gamma(k)$.

(5) $C(n)$ may also be characterized as the set of all integers h for which $\lambda(h) \mid n$.

And from (3), first with $m = \gamma(s)$ and $n = s$, then with $m = s$ and $n = \lambda(s)$, we have:

(6) For each integer s, $\lambda(\gamma(s)) \mid s$ and $s \mid \gamma(\lambda(s))$.

From (3) and the above-stated relation between λ and ϕ we see:

(7) If n is odd, $\gamma(n) = 2$; if n is even, $\lambda(n)$ is twice the least common multiple

$M(n)$ of all prime powers p^e for which $p^{e-1}(p-1)\mid n$. Hence if n is even, $24\mid n$.

The function M was introduced by Carmichael in [2], where there is a table of values of $M(n)$ for $n \leqq 150$ and such that the equation $\phi(n)=n$ has a solution. To complete this table up to $n=150$, we record the following values of $M(n)$: $M(n)=12$ if n is 14, 26, 34, 38, 62, 74, 86, 94, 98, 118, 122, 134, 142, or 146; $M(n)=120$ if n is 68, 76, or 124; $M(50)=132$; $M(114)=252$; $M(90)=4898124$.

REFERENCES

1. R. D. Carmichael, The Theory of Numbers, New York, 1914.
2. R. D. Carmichael, Notes on the simplex theory of numbers, II. An extension of Fermat's theorem, Bull. Amer. Math. Soc., vol. 15, 1908–9, pp. 221–222.

ON THE LEAST POSSIBLE ODD PERFECT NUMBER

H. A. BERNHARD, Newark College of Engineering

In 1908 A. Turčaninov proved that no odd number less than two million can be perfect [3c], a figure which is generally accepted as the minimum in standard texts [for example, see 1, 4]. However, it is easy to show by means of well-known proofs that the least possible odd perfect number is greater than ten billion.

Let the prime factorization of any odd perfect number N_0 be denoted by

$$N_0 = p_1^{a_1} p_2^{a_2} \cdots p_m^{a_m} q_1^{b_1} q_2^{b_2} \cdots q_n^{b_n}, \tag{1}$$

where $p_1, p_2, \cdots, p_m, q_1, q_2, \cdots, q_n$ are odd primes, $a_1, a_2, \cdots, a_m$ are odd integers, and $b_1, b_2, \cdots, b_n$ are even integers. It has been known for centuries [3a] that

$$m = 1, \tag{1a}$$

$$p_1 \equiv 1 \pmod 4, \tag{1b}$$

$$a_1 \equiv 1 \pmod 4. \tag{1c}$$

In addition, Sylvester [3b] has demonstrated that

$$n \geqq 4. \tag{1d}$$

Certain other conditions have been established. Steuerwald has shown

$$b_1 = b_2 = \cdots = b_n = 2 \text{ is impossible.} \tag{1e}$$

More recently, Brauer [2] extended this theme to prove that

$$b_i \neq 4, \text{ when } b_1 = b_2 = \cdots = b_{i-1} = b_{i+1} = \cdots = b_n = 2. \tag{1f}$$

Another useful fact has been demonstrated by Sylvester [5], namely, that

$$N_0 \not\equiv 0 \pmod{3\cdot 5\cdot 7}. \tag{1g}$$

The only numbers less than ten billion which satisfy conditions (1a), (1b), $\cdots$, (1g) are $3^6\cdot 5^2\cdot 11^2\cdot 13^2\cdot 17$ and $3^6\cdot 5^2\cdot 11^2\cdot 13\cdot 17^2$. Each is found to be abundant by directly computing the sum of its divisors.

References

1. W. W. R. Ball, Mathematical Recreations and Essays, American Edition, New York: Macmillan (1947), pp. 67–68.

2. A. Brauer, "On The Non-Existence of Odd Perfect Numbers of Form $p^a q_1^2 q_2^2 \cdots q_{i-1}^2 q_i^4$," Bulletin of the American Mathematical Society, vol. 49 (1943), pp. 712–718.

3a. L. E. Dickson, History of the Theory of Numbers, Washington: Carnegie (1919) vol. 1, pp. 14, 19.

3b. ———, Ibid., p. 27.

3c. ———, Ibid., p. 29.

4. O. Ore, Number Theory and Its History, First Edition, New York: McGraw-Hill (1948), pp. 91–94.

5. J. J. Sylvester, The Collected Mathematical Works of James Joseph Sylvester, Cambridge: University Press (1912) vol. 4, p. 590.

CLASSROOM NOTES

EDITED BY C. B. ALLENDOERFER, Haverford College

All material for this department should be sent to C. B. Allendoerfer, Haverford College, Haverford, Pennsylvania.

COORDINATE SYSTEMS PROJECTED ON BLACKBOARDS

C. B. ALLENDOERFER, Haverford College

It was suggested by R. M. Sutton (this MONTHLY, vol. 54, p. 276) that effective teaching use could be made of coordinate systems projected on classroom blackboards from slides or film strips. Although it may seem unlikely at first thought, the projection of white coordinate grids with black backgrounds onto blackboards gives satisfactory results even in undarkened rooms. The use of this method enables the teacher to undertake accurate plotting on a great variety of different types of coordinate systems.

For most teachers the practical difficulty of using this technique lies in the preparation of suitable slides or film strips. Through the courtesy and cooperation of a local concern a variety of such slides was prepared for use at Haverford College. In the light of our experience with them, a film strip of 36 frames was prepared and copies can be obtained from the manufacturer.

This film strip contains several varieties of rectangular cross sections, polar coordinates, logarithmic and semi-logarithmic systems, and probability and log-probability grids. The abscissa of one of the rectangular grids is calibrated in terms of fractions and multiples of π for use in plotting the graphs of trigonometric functions. Other frames contain a single straight line, parallel straight lines, and a circle. These are projected on string models of cones and other ruled surfaces to illustrate conic sections and simple space curves.

Although this film is still regarded as experimental, limited commercial production is available at reasonable cost. Inquiries should be addressed to: Engineers Publishing Company, 401 N. Broad Street, Philadelphia 8, Pa.

SUMS OF SINES CONVERTED INTO NUMERICAL SUMS

H. H. DOWNING, University of Kentucky

Interesting results are obtained from equations involving sums, both finite and infinite, of sines of multiples of x. Both members of the equation are divided by x, x is allowed to approach zero, and limits are then found.

For the first illustration let us take the trigonometric formula (Rothrock, *Elements of Plane and Spherical Trigonometry*, 1914, Example 3, page 109).

$$\sin x + \sin 2x + \cdots + \sin nx = \sin \frac{(n+1)x}{2} \cdot \sin \frac{nx}{2} \Big/ \sin \frac{x}{2} \cdot \tag{1}$$

Divide both members by x and write as follows:

$$\frac{\sin x}{x} + 2\frac{\sin 2x}{2x} + \cdots + n\frac{\sin nx}{nx}$$

$$= \frac{n+1}{2} \cdot \frac{\sin \dfrac{(n+1)x}{2}}{\dfrac{n+1}{2}x} \cdot \frac{n}{2} \cdot \frac{\sin \dfrac{nx}{2}}{\dfrac{n}{2}x} \Bigg/ \frac{1}{2} \cdot \frac{\sin \dfrac{x}{2}}{\dfrac{x}{2}} \cdot$$

Now allow x to approach zero. The resulting limit is

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2},$$

the formula for the sum of the first n consecutive integers.

As a second example consider the trigonometric formula (Rothrock, loc cit., Example 5, page 109).

$$\sin x + \sin 3x + \cdots + \sin (2n-1)x = \sin^2 nx/\sin x. \tag{2}$$

Divide by x and write in the form

$$\frac{\sin x}{x} + 3\frac{\sin 3x}{3x} + \cdots + (2n-1)\frac{\sin (2n-1)x}{(2n-1)x} = n^2 \frac{\sin^2 nx}{n^2x^2} \Big/ \frac{\sin x}{x} \cdot$$

Pass to limits and find

$$1+3+5+\cdots+(2n-1)=n^2,$$

the sum of the first n odd integers.

Next, consider the Fourier expansion (Widder, *Advanced Calculus*, 1947, Example C, page 327)

$$x = 2\left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots + (-1)^{n-1}\frac{\sin n\pi}{n} + \cdots\right), \quad -\pi \leqq x \leqq \pi. \tag{3}$$

Divide by x and obtain

$$1 = 2\left(\frac{\sin x}{x} - \frac{\sin 2x}{2x} + \frac{\sin 3x}{3x} - \cdots + (-1)^{n-1}\frac{\sin nx}{nx} + \cdots\right).$$

Let x approach zero and divide by 2. We obtain

$$1/2 = 1 - 1 + 1 - 1 + \cdots.$$

This result is found by the Cesàro summability method (Widder, loc. cit., Example A, page 263).

The next example is interesting in that it relates the preceding oscillating series and an infinite series giving $\pi/4$. Consider the Fourier expansion (Miller, *Partial Differential Equations*, 1941, Exercise 8, page 150)

$$\begin{aligned} x = {} & \left(1+\frac{2}{\pi}\right)\sin x - \frac{1}{2}\sin 2x + \left(\frac{1}{3}-\frac{2}{3^2\pi}\right)\sin 3x - \frac{1}{4}\sin 4x \\ & + \left(\frac{1}{5}+\frac{2}{5^2\pi}\right)\sin 5x - \cdots, \qquad 0 < x \leqq \pi/2. \end{aligned} \tag{4}$$

Divide by x, pass to limits, and obtain

$$1 = \left(1+\frac{2}{\pi}\right) - 1 + \left(1 - \frac{2}{3\pi}\right) - 1 + \left(1+\frac{2}{5\pi}\right) + \cdots,$$

or, on rearranging,

$$1 = (1 - 1 + 1 - 1 + \cdots) + \frac{2}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \cdots\right). \tag{5}$$

The series in the second parenthesis is the Leibnitz or Gregory series for $\pi/4$. Thus, if either the first or the second parenthesis is taken as known, the other can be determined.

ELEMENTARY PROBLEMS AND SOLUTIONS

EDITED BY HOWARD EVES, Oregon State College

Send all communications concerning Elementary Problems and Solutions to Howard Eves, Mathematics Department, Oregon State College, Corvallis, Oregon. This department welcomes problems believed to be new, and demanding no tools beyond those ordinarily furnished in the first two years of college mathematics. To facilitate their consideration, solutions should be submitted on separate, signed sheets, within three months after publication of problems.

PROBLEMS FOR SOLUTION

E 886. *Proposed by P. L. Chessin and R. Gilman, New York City*

Let M be a fixed point on the radius OR of a circle of center O and set $OR/OM=k$. If AB is a variable chord such that $\angle AMR=\angle OMB$, then (1) AB passes through a fixed point on line OR, (2) $AB=k(MB-MA)$, (3) A, B, O, M are concyclic.

E 887. *Proposed by C. W. Trigg, Los Angeles City College*

Let N be an n-digit integer and M the kn-digit integer formed by writing N down k times. Show that there is no scale of notation in which $N^k=M$ if $k>1$.

E 888. *Proposed by H. D. Grossman, New York City*

Show how to cut a hole in a cube through which another cube of equal size can pass.

E 889. *Proposed by Michael Golomb, Purdue University*

Let k be a circle with its interior, F_1F_2 one of its diameters, P a point on the axis of the circle, F_1PF_2 a circular arc, and A_1PA_2 a semiellipse with foci at F_1, F_2. Show that the solid angles subtended by k at the points of arc F_1PF_2 are no smaller, and at the points of arc A_1PA_2 are no greater, than the solid angle subtended by k at P.

E 890. *Proposed by Leo Moser, University of Manitoba*

Let $a_1, \cdots, a_n$ be n integers with $a_1=1$ and $a_i\leqq a_{i+1}\leqq 2a_i$, $i=1, \cdots, n-1$. Show that there exists a sequence $\{\epsilon_i\}$ of plus and minus ones such that $\sum_{i=1}^n\epsilon_i a_i=0$ or 1.

SOLUTIONS

Pythagorean Triangles with Equal Perimeters

E 812 [1948, 248 and 1949, 32]. *Proposed by Monte Dernham, San Francisco, California*

Find the shortest perimeter common to two different primitive Pythagorean triangles.

Comment by C. W. Trigg, Los Angeles City College. There are shorter perimeters common to two and to three non-primitive Pythagorean triangles than those reported by Ogilvy and Buker in this MONTHLY [1949, 33]. The shortest perimeters common to x non-primitive Pythagorean triangles, together with the corresponding sets of triangles, are given below.

$x=2$. $p=60$; (15, 20, 25), (10, 24, 26).
$x=3$. $p=120$; (30, 40, 50), (20, 48, 52), (45, 24, 51).
$x=4$. $p=360$; (90, 120, 150), (60, 144, 156), (135, 72, 153), (36, 160, 164).
$x=5$. $p=660$; (165, 220, 275), (110, 264, 286), (55, 300, 305), (297, 60, 303), (210, 176, 274).
$x=6$. $p=720$; (180, 240, 300), (270, 144, 306), (315, 80, 325), (45, 336, 339), (72, 320, 328), (120, 288, 312).

Some sets of triangles, containing one primitive member, have a common perimeter shorter than those listed above. That is,

$x=4$. $p=240$; (60, 80, 100), (40, 96, 104), (90, 48, 102), (15, 112, 113).
$x=5$. $p=420$; (105, 140, 175), (70, 168, 182), (175, 60, 185), (126, 120, 174), (195, 28, 197).
$x=7$. $p=1320$; (330, 440, 550), (220, 528, 572), (495, 264, 561), (110, 600, 610), (594, 120, 606), (430, 352, 548), (231, 520, 569).
$x=8$. $p=840$; (210, 280, 350), (140, 336, 364), (315, 168, 357), (105, 360, 375), (252, 240, 348), (350, 120, 370), (390, 56, 394), (399, 40, 401).
$x=10$. $p=1680$; (420, 560, 700), (280, 672, 728), (630, 336, 714), (210, 720, 750), (504, 480, 696), (700, 240, 740), (105, 784, 791), (780, 112, 788), (798, 80, 802), (455, 528, 697).

In addition, we find that the pair of Pythagorean triangles having the smallest common perimeter and a common side is (21, 28, 35) and (35, 12, 37), $p=84$.

A Series for π

E 854 [1949, 104]. *Proposed by Jerome C. R. Li, Oregon State College*

Show that $\pi=\sum_{n=0}^{\infty}(n!)^2 2^{n+1}/(2n+1)!$.

I. *Solution by Ragnar Dybvik, Levanger, Norway.* (Using the beta function.) We have

$$\sum_{n=0}^{\infty}(n!)^2 2^{n+1}/(2n+1)! = \sum_{n=0}^{\infty} 2^{n+1}\Gamma(n+1)\Gamma(n+1)/\Gamma(2n+2)$$

$$= \sum_{n=0}^{\infty} 2^{n+1}B(n+1, n+1) = \sum_{n=0}^{\infty} 2^{n+1}\int_0^1 x^n(1-x)^n dx$$

$$= 2\int_0^1 \sum_{n=0}^{\infty}(2x)^n(1-x)^n dx = 2\int_0^1 [1-2x(1-x)]^{-1}dx$$

$$= \int_0^1 [(x-1/2)^2+(1/2)^2]^{-1}dx = 2\arctan 2(x-1/2)\Big|_0^1 = \pi.$$

II. *Solution by Paul Carnahan, University of Texas.* (Using Wallis's formula.) From Wallis's formula for an odd power,

$$\int_0^{\pi/2} \cos^{2n+1} x dx = (n!)^2 2^{2n}/(2n+1)!.$$

Therefore

$$(n!)^2 2^{n+1}/(2n+1)! = 2\int_0^{\pi/2} 2^{-n} \cos^{2n+1} x dx$$
$$= 2\int_0^{\pi/2} \cos x(\tfrac{1}{2}\cos^2 x)^n dx,$$

and

$$\sum_{n=0}^{\infty} (n!)^2 2^{n+1}/(2n+1)! = 2\int_0^{\pi/2} \cos x\left[\sum_{n=0}^{\infty} (\tfrac{1}{2}\cos^2 x)^n\right] dx$$
$$= \int_0^{\pi/2} \cos x(1 - \tfrac{1}{2}\cos^2 x)^{-1} dx = \pi.$$

III. *Solution by M. S. Klamkin, Polytechnic Institute of Brooklyn.* (Using the Legendre polynomials.) We have

$$a_n = (n!)^2 2^{n+1}/(2n+1)! = \int_{-1}^{1} z^n P_n(z) dz,$$

where $P_n(z)$ is the Legendre polynomial of degree n. Therefore

$$\sum_{n=0}^{\infty} a_n = \int_{-1}^{1} \sum_{n=0}^{\infty} z^n P_n(z) dz = \int_{-1}^{1} (1 - z^2)^{-1/2} dz = \pi.$$

IV. *Solution by N. J. Fine, University of Pennsylvania.* (Using a differential equation.) For $|z| < 4$, define

$$F(z) = \sum_{n=0}^{\infty} \binom{2n+1}{n}^{-1} z^{n+1}/(n+1).$$

It is easy to show that $F(z)$ satisfies the differential equation

$$(4 - z)F'(z) - 2F(z)/z = 2.$$

Setting $z = 4\sin^2\alpha$, $0 \leq \alpha < \pi/2$, and $y(\alpha) = F(z)$, the differential equation for $y(\alpha)$ is found to be

$$d(y \cot \alpha) d\alpha = 4.$$

Therefore $y = (4\alpha + C)\tan \alpha$. Observing that $y(\alpha)$ has a double zero at $\alpha = 0$, we find that $C = 0$. Thus, setting $\alpha = \pi/4$, we find $F(2) = y(\pi/4) = \pi$, which is the desired result.

V. *Solution by D. H. Browne, Buffalo, N. Y.* (Using Euler's transformation of series.) If we apply Euler's transformation

$$\sum_{n=0}^{\infty}(-1)^n v_n = \sum_{n=0}^{\infty}(\Delta^n v_0)/2^{n+1}$$

to the Leibnitz series

$$\pi/4 = \sum_{n=0}^{\infty}(-1)^n/(2n+1)$$

we find

$$\Delta^n v_0 = (n!)^2 2^{2n}/(2n+1)!,$$

whence

$$\pi = \sum_{n=0}^{\infty}(n!)^2 2^{n+1}/(2n+1)!.$$

Also solved by Louis Berkofsky, W. G. Brady, Roger Lessard, and E. D. Rainville.

Solution V may be found in Bromwich, *Theory of Infinite Series*, 2nd ed., p. 62; Knopp, *Theory and Application of Infinite Series*, p. 244 and ex. 2, p. 246; Chrystal, *Text Book of Algebra*, part II, pp. 408–409.

Four Cospherical Circles

E 855 [1949, 104]. *Proposed by Victor Thébault, Tennie, Sarthe, France*

Planes through the orthocenter of an orthocentric tetrahedron perpendicular to four concurrent cevians cut the spheres described on these cevians in four cospherical circles.

Solution by the Proposer. Let $ABCD$ be the tetrahedron, AP_a, AP_b, AP_c, AP_d the four cevians concurrent in P, and H the orthocenter. Let us designate the four circles by (C_a), (C_b), (C_c), (C_d), and let A', B', C', D' be the feet of the altitudes of the tetrahedron. The sphere on AP_a as diameter passes through A'. Therefore the power of H with respect to circle (C_a) is given by $(HA)(HA')$. Now

$$(1)\qquad \begin{aligned}(HA)(HA') &= (HB)(HB') = (HC)(HC') = (HD)(HD')\\ &= (HO^2 - R^2)/3 = \rho^2,\end{aligned}$$

where O is the circumcenter, and R and ρ are the radii of the circumsphere and the conjugate sphere of $ABCD$. Therefore H has the same power with respect to each of the four circles (C_a), (C_b), (C_c), (C_d). Since the normals to the planes of these circles at their centers are concurrent at P, this is the center of a sphere

(S_1) passing through the four circles. The square of the radius of this sphere is, then,

$$R_1^2 = HP^2 - \rho^2 = HP^2 - R^2 + (a^2 + a'^2)/4, \tag{2}$$

where a and a' are any pair of opposite edges of $ABCD$.

Editorial Note. For relations (1) and (2) see N. A. Court, *Modern Pure Solid Geometry*, sections 218, 795, 823, 825.

The analogous theorem in the plane is also true, and reads as follows: Lines through the orthocenter of a triangle perpendicular to three concurrent cevians cut the circles described on these cevians as diameters in six concyclic points.

A Cyclic Number

E 856 [1949, 179]. *Proposed by J. T. Hurt, Texas Agricultural and Mechanical College*

Let N be an integer of p digits. If the last digit is removed and placed before the remaining $p-1$ digits, a new number of p digits is formed which is $(1/n)$th of the original number. Find the most general such number N.

Solution by N. D. Lane, St. Andrew's College, Ontario. Let

$$N = a_{p-1}10^{p-1} + \cdots + a_1 10 + a_0, \qquad a_{p-1} \neq 0,$$

and

$$M = a_0 10^{p-1} + a_{p-1}10^{p-2} + \cdots + a_1, \qquad a_0 \neq 0,$$

such that $M = N/n$. Since $10M = N + a_0(10^p - 1)$, we find

$$N = n(10^p - 1)a_0/(10 - n).$$

We see that $n<10$, and since 10^p-1 is the largest p-digit number, $na_0/(10-n) \leqq 1$, or $n(a_0+1) \leqq 10$, and $n \leqq 5$. The requirement that M be integral eliminates $n=5, 4, 2$. If $n=1$ we get nine solutions for each value of p. If $n=3$, a_0 must be 1 or 2, and there exist solutions to the problem if p is a multiple of 6. In this case

$$N = 3(10^{6k} - 1)/7 \quad \text{or} \quad N = 6(10^{6k} - 1)/7,$$

the two smallest such numbers being 428571 and 857142.

Also solved by Monte Dernham, B. B. Dressler, R. T. Hood, M. S. Klamkin, Roger Lessard, Azriel Rosenfeld, C. M. Sandwick, and the proposer.

ADVANCED PROBLEMS AND SOLUTIONS

EDITED BY E. P. STARKE, Rutgers University

Send all communications concerning Advanced Problems and Solutions to E. P. Starke, Rutgers University, New Brunswick, New Jersey. All manuscripts should be typewritten with double spacing and margins at least one inch wide. Problems containing results believed to be new or extensions of old results are especially sought. Proposers of problems should also enclose any solutions or information that will assist the editor. In general, problems in well known text books or results in readily accessible sources should not be proposed for this department.

PROBLEMS FOR SOLUTION

4365. *Proposed by Paul Erdös, Syracuse University*

Let $a_1 < a_2 < \cdots < a_k \leqq n$ be such that the least common multiple of any two a's exceeds n. Prove that

$$\sum_{i=1}^{k} \frac{1}{a_i} < 2.$$

4366. *Proposed by J. Rosenbaum, the Milford School, Connecticut*

Determine the condition which two concentric spheres must satisfy in order that a tetrahedron can be simultaneously inscribed in one and circumscribed about the other. Give a construction for the tetrahedron.

4367. *Proposed by F. E. Wood, University of Oregon*

Evaluate the determinant

$$\begin{vmatrix} {}_nP_n & {}_nC_1 & {}_nC_2 & \cdots & {}_nC_n \\ {}_{n-1}P_{n-1} & {}_{n-1}C_0 & {}_{n-1}C_1 & \cdots & {}_{n-1}C_{n-1} \\ {}_{n-2}P_{n-2} & 0 & {}_{n-2}C_0 & \cdots & {}_{n-2}C_{n-2} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ {}_0P_0 & 0 & \cdots & 0 & {}_0C_0 \end{vmatrix}$$

of the $(n+1)$th order where ${}_nP_r$ and ${}_nC_r$ are the usual permutation and combination symbols.

4368. *Proposed by Victor Thébault, Tennie, Sarthe, France*

In a tetrahedron $T \equiv ABCD$, the planes tangent at A, B, C, D to the circumsphere of T cut the planes of the opposite faces in four lines. A necessary and sufficient condition for these four lines to be rulings of a hyperbolic paraboloid is that T be orthocentric, and a necessary and sufficient condition for these four lines to be coplanar is that T be isodynamic.

4369. *Proposed by Orrin Frink, Pennsylvania State College*

Show that in every neighborhood of a point on a surface $z = f(x, y)$, with

$f(x, y)$ a continuous function, there exist four points on the surface which are the corners of a square.

SOLUTIONS

Metaharmonic Tetrahedrons

4228 [1946, 594]. *Proposed by Victor Thébault, Tennie, Sarthe, France*

In a tetrahedron $ABCD$ the straight lines joining each vertex to the points of intersection V and W of the six spheres of similitude of four given spheres with centers A, B, C, D, meet the circumsphere in the vertices of two equal tetrahedrons $A'B'C'D'$ and $A''B''C''D''$.

*Solution by the Proposer.** In a tetrahedron $T \equiv ABCD$, whose circumsphere is (O, R), we designate the lengths of the edges BC, DA, CA, DB, AB, DC by a, a', b, b', c, c'. Let V and W be the points common to the six spheres of similitude of the four given spheres (A, l), (B, m), (C, n), (D, p). Then the specified points A', B', C', D' and A'', B'', C'', D'' are transforms of the vertices A, B, C, D of T by inversions of poles V and W and of powers

$$\lambda^2 = \overline{VO}^2 - R^2 \quad \text{and} \quad \mu^2 = \overline{WO}^2 - R^2.$$

The theory of inversion shows that

$$B'C' = \lambda^2 a/VB \cdot VC, \qquad D'A' = \lambda^2 a'/VD.VA, \cdots,$$
$$B''C'' = \mu^2 a/WB \cdot WC, \qquad D''A'' = \mu^2 a'/WD \cdot WA, \cdots.$$

Hence

$$B'C'/D'A' = a \cdot VD \cdot VA/a' \cdot VB \cdot VC = a \cdot WD \cdot WA/a' \cdot WB \cdot WC, \cdots.$$

But $VA:VB:VC:VD = l:m:n:p = WA:WB:WC:WD$, so that

$$B'C'/D'A' = alp/a'mn = B''C''/D''A'',$$
$$C'A'/D'B' = bmp/b'ln = C''A''/D''B'',$$
$$A'B'/D'C' = cnp/c'lm = A''B''/D''C''.$$

The metaharmonic tetrahedrons $A'B'C'D'$ and $A''B''C''D''$ of T, associated with the points V and W, are similar since corresponding edges are proportional, As they are inscribed in the same sphere (O, R) they are equal.

Note. If the squares l^2, m^2, n^2, p^2 of the radii of the spheres of centers A, B, C, D are proportional to the products $a'bc$, $ab'c$, abc', $a'b'c'$ of the edges of T, we have shown that the corresponding metaharmonic tetrahedrons are isosceles. (*Annales de la Société Scientifique de Bruxelles*, 1947, p. 15.)

---

* Translated by W. E. Byrne, Virginia Military Institute.

Lower Bound to a Trigonometric Sum

4290 [1948, 253]. *Proposed by P. T. Bateman, Yale University*

The function $-\log |2 \sin \frac{1}{2}x|$ has the Fourier series

$$\cos x + \frac{1}{2} \cos 2x + \frac{1}{3} \cos 3x + \cdots.$$

Prove that no partial sum of the series is ever less than -1.

Solution by Otto Szasz, University of Cincinniat. Let

$$S_n(x) = \sum_{v=1}^{n} \frac{\cos vx}{v} \geqq -1, \qquad n = 1, 2, \cdots; 0 \leqq x \leqq \pi.$$

We have

$$-S_n'(x) = \sum_{v=1}^{n} \sin vx = \left\{\cos \tfrac{1}{2}x - \cos (n + \tfrac{1}{2})x\right\}/2 \sin \tfrac{1}{2}x.$$

Hence

$$-S_{n-1}'(x) - S_n'(x) = (1 - \cos nx)/\tan \tfrac{1}{2}x \geqq 0, \qquad n \geqq 2; 0 \leqq x \leqq \pi.$$

It follows that $-S_{n-1}(x)-S_n(x)$ increases as x increases to π, whence

$$-S_{n-1}(x) - S_n(x) \leqq -S_{n-1}(\pi) - S_n(\pi) = 2\sum_{v=1}^{n-1} (-1)^{v-1}/v + (-1)^{n-1}/n.$$

Let $n=2k$, $k \geqq 1$. Then

$$-S_{2k-1}(x) - S_{2k}(x) = -2S_{2k}(x) + \frac{\cos 2kx}{2k} = -2S_{2k-1}(x) - \frac{\cos 2kx}{2k}$$

$$\leqq 2 \sum_{v=1}^{2k-1} (-1)^{v-1}/v - 1/2k = 2t_k - 1/2k, \quad \text{say:}$$

hence

$$-2S_{2k}(x) \leqq 2t_k - (1 + \cos 2kx)/2k \leqq 2t_k,$$

and

$$-2S_{2k-1}(x) \leqq 2t_k - (1 - \cos 2kx)/2k \leqq 2t_k.$$

Obviously $t_1 = 1$, and $t_k < 1$ for $k > 1$; it follows that

$$S_n(x) \geqq -t_k \geqq -1, \qquad n \geqq 1.$$

It is easily seen that equality here holds only for $n=1$ and $x=\pi$. Furthermore t_k decreases to log 2, as $n \to \infty$, and

$$\liminf_{n\to\infty} \min_{0\leqq x\leqq \pi} S_n(x) = -\log 2.$$

Also solved by R. P. Boas, Jr., Fritz Herzog, E. Lukacs, C. D. Olds, Norman Miller, and the Proposer.

Editorial Note. The present result is due to W. H. Young (*Proceedings of the London Mathematical Society*; v. 11, 1913, p. 359) and has been here reproposed because of an error in Young's proof. The problem is also found in Polya-Szegö, *Aufgaben und Lehrsätze aus der Analysis*, v. 2, p. 79, problem 28, the solution being given on p. 271. In this solution Young's error is corrected.

An Enumeration Problem

4291 [1948, 253]. *Proposed by J. P. Ballantine, University of Washington, Seattle*

$$\begin{array}{cccccc} 1 & 2 & 3 & \cdots & n \\ 0 & 1 & 2 & \cdots & n-1 \\ 0 & 0 & 1 & \cdots & n-2 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdots & 1 \end{array}$$

Consider the square array with 1's down the principal diagonal, increasing consecutive integers above the diagonal, and zeros below. (a) How many different paths of $i+j-2$ steps are possible from the 1 in the upper left corner to the element in the ith row and jth column, $j \geqq i$, without passing through any zero element?

(b) Define the value of each path as the product of the elements through which the path passes, not counting the terminal elements. Show that the sum of the values of all the paths in (a) is the coefficient of $(\tan x)^{j-i+1}$ in the $(j+i-2)$nd derivative of $\tan x$ with respect to x, expressed in terms of $\tan x$.

Solution by Fritz Herzog, Michigan State College. Let $i \leqq j$ and denote by (i, j) the element $j-i+1$ in the ith row and jth column of the given array. We call a path from $(1, 1)$ to (i, j) a forbidden path if it passes through at least one zero element, otherwise it is a proper path. Let the number of proper paths required in (a) be represented by $N(i, j)$. Evidently $N(i, j) = N(i, j-1) + N(i-1, j)$, a relation which leads by an easy induction to the result below. The following alternate derivation has some interest.

Let P be any forbidden path and let $(k, k-1)$ be the first zero element reached by P. $(2 \leqq k \leqq i.)$ Let $P = P_1 + P_2$, where P_1 is the part of P from $(1, 1)$ to $(k, k-1)$ and P_2 the part of P from $(k, k-1)$ to (i, j). Let Q_1 be the reflection of P_1 about the principal diagonal, so that Q_1 is a path from $(1, 1)$ to $(k-1, k)$, and let Q_2 be obtained by translating P_2 so as to begin at $(k-1, k)$ and thus end at $(i-1, j+1)$. $Q = Q_1 + Q_2$ is then a path (not necessarily a proper path) from $(1, 1)$ to $(i-1, j+1)$. If P' and P'' are two different forbidden paths from $(1, 1)$ to (i, j) then the first unit segment of P' (P'') which is not common to P' and P'' will belong either to P_1' (P_1'') or to P_2' (P_2''). This shows that the corresponding paths

Q' and Q'' will also be different from one another. Finally, every path Q from $(1, 1)$ to $(i-1, j+1)$ is actually obtained from a forbidden path P; in order to construct this path P, we need only define $(k-1, k)$ as the first element to the right of the principal diagonal reached by Q and then reverse the process described above. Thus a one-to-one correspondence has been established between the forbidden paths from $(1, 1)$ to (i, j) and all paths from $(1, 1)$ to $(i-1, j+1)$. The number of the latter is known to equal the binomial coefficient ${}_{j+i-2}C_{i-2}$, while the number of all paths from $(1, 1)$ to (i, j) equals ${}_{j+i-2}C_{i-1}$. The answer to (a) is therefore

$$N(i, j) = {}_{j+i-2}C_{i-1} - {}_{j+i-2}C_{i-2} = \frac{(j - i + 1)(j + i - 2)!}{(i - 1)!j!}.$$

Let the sum of all the values of all the paths (not necessarily proper) from $(1, 1)$ to (i, j) be denoted by $S(i, j)$. Any path ending at (i, j), $i>1, j>1$, must pass through either $(i-1, j)$ or $(i, j-1)$. Consequently, we have

$$(1) \qquad S(i, j) = (j - i + 2)S(i - 1, j) + (j - i)S(i, j - 1),$$

and this relation will hold for $1 \leqq i \leqq j+1$, $(i, j) \neq (1, 1)$, if we define $S(0, r) = S(r, r-2) = 0$ for $r \geqq 2$ and $S(1, 1) = 1$. (As a consequence $S(1, 2) = S(2, 1) = 1$. These values, although not defined by the proposal, are acceptable in the same sense as $0! = 1$.)

We shall show that

$$(2) \qquad D_x^n(\tan x) = \sum_{k=0}^{[(n+1)/2]} S(k + 1, n - k + 1)(\tan x)^{n-2k+1}, \qquad n = 0, 1, 2, \cdots.$$

Because of $S(1, 1) = 1$, (2) is true for $n = 0$. Assume the equation

$$D_x^{n-1}(\tan x) = \sum_{k=0}^{[n/2]} S(k + 1, n - k)(\tan x)^{n-2k}$$

has been proved for a positive integer n. Differentiating and making use of $S(0, r) = S(r, r-2) = 0$, $r \geqq 2$, we obtain

$$\begin{aligned} D_x^n(\tan x) &= \sum_{k=0}^{[(n-1)/2]} (n - 2k)S(k + 1, n - k)(\tan x)^{n-2k-1}(1 + \tan^2 x) \\ &= \sum_{k=0}^{[(n+1)/2]} \{(n - 2k + 2)S(k, n - k + 1) \\ &\qquad + (n - 2k)S(k + 1, n - k)\}(\tan x)^{n-2k+1}. \end{aligned}$$

Applying (1) to the expression in braces, we obtain (2), which is thus established by mathematical induction.

Substituting $k+1 = i$, $n-k+1 = j$ in (2), we conclude that $S(i, j)$ is the coefficient of $(\tan x)^{j-i+1}$ in the $(j+i-2)$nd derivative of $\tan x$.

Also solved by Y. S. Luan, and the Proposer.

RECENT PUBLICATIONS

EDITED BY E. P. VANCE, Oberlin College

All books for review should be sent directly to the editor of this department, American Mathematical Monthly, 531 West 116th Street, New York 27, N. Y. and not to any of the other editors or officers of the Association.

Number Theory and Its History. By Oystein Ore. New York, McGraw-Hill Book Co., Inc., 1948. x+370 pages. $4.50.

This book gives an interesting account of many topics of elementary number theory, interwoven with considerable historical material. The theory is available to readers with a limited mathematical knowledge, since the mathematical requirements are reduced to a minimum, and only "topics of systematic and historical importance capable of a simple presentation" are chosen. The author has succeeded in making the book attractive to amateurs, to people interested in number theory as a pastime; and it is well suited as a text for a first course in number theory especially for students who may not wish to study the subject further.

For students looking to further use of the subject, the desirability of a mixed mathematical-historical development may well be debated. While the present treatment is interesting, and hence probably inspiring, it has the disadvantage that the most significant ideas and methods do not stand out sufficiently. A teacher well acquainted with the subject could make up for this deficiency.

The historical material seems to be largely up to date. The history of number symbols associated with various cultures is given at length in Chapter 1, with plates of finger symbols, tallies, Egyptian, Greek, Chinese, and other numerals. Reproductions are given in Chapter 6 of part of the Papyrus Rhind, and in Chapter 8 of two Babylonian tablets, giving tables of reciprocals and of right triangles with integral sides. Early mathematical writers are frequently quoted; problems proposed by them are given, and their methods of solution are contrasted with modern methods. Most of the mathematical sections are accompanied by remarks on recent developments on those topics.

Among the mathematical sections may be listed divisibility of numbers (including binary number systems), Euclid's algorithm, prime numbers (including a discussion of their distribution), the simplest arithmetical functions, an extensive discussion of linear indeterminate equations, Diophantine problems, congruences, Wilson's and Euler's theorems and their consequences, theory of decimal expansions, classical construction problems. It is regrettable that no mention is made of quadratic residues (or hence of Gauss's "gem of the higher arithmetic," the quadratic reciprocity law). Among topics not usually found in textbooks there may be noted Thue's theorem that if a is prime to p, $ax \equiv \pm y \pmod{p}$ is solvable with x and y between 1 and $\sqrt{p}$, this being used to prove that a prime p of the form $4n+1$ is a sum of two squares. The function $\lambda(n)$ which is the least exponent such that $a^{\lambda(n)} \equiv 1 \pmod{n}$ for all a prime to n, is studied and applied to a problem in telephone splicing. The author takes the opportunity afforded by the properties of greatest common divisors, least common multiples, and of congruences, to discuss concepts of importance in mathematics generally, such as lattices, rings, and moduls.

Numerous numerical examples and historical remarks make the progress of the mathematics somewhat slow, but this may be an advantage for some readers.

The following corrections should be made. It would be desirable, on page 29, to define "a is a divisor of c" even when $c=0$, since such cases are used later. On pages 30–32, b is treated several times as though it were positive, although it was introduced as merely not zero. The greatest common divisor (p. 40) should be defined for integers not all zero. The word "number" is overworked; perhaps, integer, positive integer, nonzero integer, and so on, might be clearer in several places. Near the bottom of page 171, read inverses for inversions. On page 168, (8–5) is the general primitive solution of $a^2+b^2=c^2$ only with a even; and (8–6) is not the general solution in rationals (as stated), since in its derivation it was assumed that $v\neq 0$. Indeed, the solution $a=0$, $b=1=c$ cannot be obtained with r and t rational.

GORDON PALL

A Concise History of Mathematics. Two volumes. (Dover Series in Mathematics and Physics). By D. J. Struik. New York, Dover Publications, 1948. 18+299 pages. $1.50 per volume.

To write on the history of mathematics has been, from old times to our days, a very attractive task. To write a history of mathematics is a tremendous undertaking because it is identical with a presentation of mathematics itself immersed in its human surroundings. As an approximate completeness is here even more unthinkable than in other historic fields, it is laudable to write a concise history as this is presented.

This work has the great advantage of being written by an accomplished mathematician as well as a reliable historian. Of course, the choice of what is described and what is omitted depends on individual taste. In the beginning we find, caused by recent discoveries about pre-Greek mathematics, a relatively detailed report on the beginnings. The report on the nineteenth century, on the other hand, is obviously influenced by F. Klein's inspiring work on the development of mathematics during this period.

The reviewer regrets that the author does not mention the beginning of Axiomatics and does not give the name of the basic work of the "Father of Axiomatics," M. Pasch. This trend in the last quarter of the nineteenth century is important because of its continuation in the present century, particularly in this country.

There are, of course, many such points, more or less important, with which some people might disagree. However, this cannot change the impression that it is a very valuable book, especially for three reasons. First, the reader will find many interesting details, difficult to find in other, much more voluminous treatises. Second, the author gives ample text references. Also, there are many portraits, drawings and text reproductions. It is only regrettable that, apparently, it was not possible to print the reproductions on a more adequate paper.

M. DEHN

A Philosophy of Mathematics. By L. O. Kattsoff. Ames, Iowa, Iowa State College Press, 1948. 9+266 pages. $5.00.

The aim of this book is to "give the beginning student an introduction to the many problems raised by the queen of the sciences." The author considers that existing treatments of the subject are either too difficult or else not sufficiently comprehensive for this purpose. The book is intended for students whose knowledge of mathematics and philosophy is that of an undergraduate senior. The author is a philosopher, and the work has naturally a primarily philosophical cast.

The book begins with a discussion of the definition of mathematics and its objects. It then considers the notion of natural number, with emphasis on the various ways of introducing it. The next topic is the extensions of the number system to rational, real, and complex numbers. This leads to an elementary discussion of the theory of abstract sets and transfinite numbers. An exposition of the paradoxes then follows. The next four chapters are devoted to the various methods of founding a system of mathematics, viz.: the logistic, formalistic, elementalistic (*i.e.* the theory of Church), and the intuitionistic foundations. Then there is a chapter on the Gödel theorem and its significance; and one on the "signific" standpoint of Mannoury. Finally there are three chapters devoted to an analysis of mathematical symbolism, of the methods of mathematical proof, of various sorts of definitions, of postulational procedures, and of the relations of mathematics to reality.

The treatment is rather sweeping in scope. Among the topics fitted into the above outline are the following: the notion of magnitude according to Bolzano; the propositional algebra and predicate calculus of the *Principia Mathematica*, as well as the definitions relating to cardinal number; Frege's symbolism; the "empirico-postulational" definition of Pasch; an axiom set for the natural numbers due to Neder, but not the original system of Peano; the denumerability of certain aggregates and the diagonal process; the theory of types and the axiom of reducibility; the theories of Chwistek; Heyting's formalization of intuitionistic mathematics, and the main ideas of intuitionism; and a brief treatment of modal logic. Several of these topics are somewhat unusual for a book of this kind. Moreover the book goes into great detail on certain topics, such as the Church theory of metads, which are hardly suited to its purpose, both on the ground that they are highly difficult and technical and because their importance is somewhat controversial. On the other hand there are, as is naturally to be expected, several topics which the reviewer thinks are slighted. Among those not mentioned at all, which nevertheless seem of major importance for the purpose of such a book, are the theory of recursive functions, the rule-theoretic methods of Gentzen, the stratification methods of Quine, and the semiotical methods of Carnap.

The reviewer finds the style of the book exceedingly obscure. This obscurity is coupled with carelessness in regard to matters of detail. The book contains a number of sloppy mistakes of which the definition of the sum of two cardinals on page 84 is typical: the author states that if m is the cardinal number of a class M, and n is the cardinal number of N, then $m+n$ is the cardinal number of $M+N$, without the obvious restriction that M and N be mutually exclusive. The reader who wishes to see the extent and gravity of these errors should read the able review by Canon Feys in the *Journal of Symbolic Logic*, vol. 13, pp. 208–212 (1948). Certainly the book is not to be recommended to any mathematician, young or old, who is seeking information on the foundations of mathematics, and is not in a position to have a critical judgment of his own.

H. B. CURRY

Modern Operational Calculus. By N. W. McLachlan. Cambridge, at the University Press; New York, The Macmillan Company, 1948. 12+218 pages, 29 illustrations. $5.00.

This book presents a concise treatment of the Laplace transform method of solution of ordinary and partial differential equations. It is illustrated by typical examples taken from electric circuits, electric transmission lines, heat flow, acoustics, and so on. The book is intended primarily for postgraduate engineers. Laplace transforms are defined in Chapter 1 and special rules for their formation are derived in Chapter 2. Chapter 3 explains the use of Laplace transforms in solving ordinary differential equations, and Chapter 4 deals with partial differential equations. These four chapters account for one-half of the book; the rest is devoted to evaluation of integrals, derivation of Laplace transforms, and appendices containing various definitions and theorems pertaining to infinite series and infinite integrals. At the end of the book there is a large collection of problems and a short table of Laplace transforms.

The author has included a great deal of useful information in this small volume and many engineers will find it a convenient reference book.

It should be mentioned however that the author uses the "p-multiplied transform" defined by

$$\phi(p) = p\int_0^\infty \epsilon^{-pt}f(t)dt$$

instead of the more conventional transform

$$F(p) = \int_0^\infty \epsilon^{-pt}f(t)dt.$$

While it is easy to change from one type of transform to the other, the difference must constantly be kept in mind when referring to other books and tables of transforms. The author's reasons for the change of definition are two: (1) to make Laplace transforms identical with Heaviside's operational forms, and (2) to obtain the dimensional equivalence of time functions and their Laplace transforms. The price of these two items is the loss of the simple interpretation of the conventional transform as the relative amplitude of a typical frequency component of the spectrum of the corresponding time function.

S. A. SCHELKUNOFF

Cours de Mécanique Rationelle. Third Edition. By Jean Chazy. Paris, Gauthier-Villars, 1947–8. Tome I, *Dynamique du Point Matériel*, 1947. 5+482 pages. 900 francs. Tome II, *Dynamique des Systèmes Matériels*, 1948. 6+511 pages. 1100 francs.

Professor Chazy's *Cours* provides another lucid, and in this case refreshing, introduction to theoretical dynamics. The work now appears in the third edition. The two volumes first appeared in 1933 and were reviewed by Professor W. R. Longley in the *Bulletin*, *39*, 491, 1933 and *41*, 13, 1935, respectively.

The first volume is devoted to particle dynamics and the second to the dynamics of rigid bodies and dynamical systems in general. Together they comprise the course of lectures which the author gave at the Faculté des Sciences de Paris from 1928 to 1941, and, excepting kinematics, represent the program for the *Certificat de Mécanique Rationnelle.* Explicitly recognizing the need for exercises, which seem to be customarily omitted from French treatises on mechanics, Professor Chazy has again inserted the lists of the *Certificat* questions, now, in seventy pages, for the years 1928 to 1946. This collection is appended to the first volume, but the questions themselves in large part call for a knowledge of the material which is reserved to the second volume.

The subject matter of the second volume is largely the traditional one, together with chapters on hydrostatics, hydrodynamics, and the elements of the theory of Newtonian attraction and potential. There is, among other things, an instructive chapter on impulsive motion. Variational principles are neglected, although Hamilton's principle is briefly considered. But for the reader who wishes an account of these matters there is Professor Chazy's own treatment in the first chapter of his admirable *La Théorie de la Relativité et la Mécanique Céleste* (Paris, 1928–30).

Professor Longley (*loc. cit.*) dealt sharply with the second chapter on "the principles of mechanics" in the first volume of the present work. Somewhere one must choose a fundamental reference frame of rest in order to get on with the subject. Professor Chazy places the origin at "the center of gravity of the solar system," the direction of axes being fixed with respect to the "fixed stars." These he calls the "axes of Copernicus," and a coordinate system in uniform rectilinear translation with respect to these defines the "axes of Galileo"; motion with respect to either of these reference frames is called "absolute." An axiomatic treatment is attempted: (1) "the principle of inertia or the principle of Kepler" ("An isolated particle has uniform rectilinear motion, or zero acceleration"); (2) "the principle or axiom of Galileo or of initial conditions"; (3) "the principle of the equality of action and reaction"; (4) "the principle of the composition of accelerations and of forces" (parallelogram law). The second of these caused Professor Longley undue concern. It was Galileo's discovery that it is acceleration and not velocity (as Aristotle and the Scholastics had taught) which requires a force to initiate and maintain it. In other words the fundamental equations of motion are ordinary differential equations of the *second* order, with compatible boundary conditions. It is this vital fact which Professor Chazy wishes to emphasize, together with the consideration of the existence and uniqueness of the solution of the differential system. In the course of this "rationalizing" the author for some inexplicable reason never chooses to mention the name of Newton, but that is another matter. Shortly the motion of a particle on the earth's surface is obtained relative to the rotating earth, the centrifugal and Coriolis forces being introduced at once.

Many instructors in mechanics will prefer a different approach to the fundamental concepts of mechanics. At least the attention of promising students may be called to this book, and they might advisedly be invited to discuss critically and elaborate on Professor Chazy's stimulating account of the matter, in class or in their mathematics club.

S. G. HACKER

NEW BOOKS RECEIVED

Les Grands Courants de la Pensee Mathematique. Introduction by F. le Lionnais. Cahiers du Sud, 1948. 533 pp. 840 francs.

Calculus. 2nd Edition. By L. M. Kells. New York, Prentice-Hall, 1949. 12+508 pp. $4.00.

A Short Course in Differential Equations. By E. D. Rainville. New York, Macmillan, 1949. 10+210 pp. $3.00.

Analytic Geometry. By J. J. Corliss, I. K. Feinstein and H. S. Levin. New York, Harper, 1949. 14+370 pp. $3.25.

Plane and Spherical Trigonometry With Tables. 3rd Edition. By J. Shibli. Boston, Ginn and Company, 1949, 12+94 pp. $3.00.

College Algebra. 3rd Edition. By J. B. Rosenbach and E. A. Whitman. Boston, Ginn and Company, 1949. 10+523+42 pp. $3.00.

Table of Sines and Cosines to Fifteen Decimal Places at Hundredths of a Degree. (Applied Mathematics Series, no. 5). Prepared by the Computation Laboratory of the National Bureau of Standards. Washington, D. C., U. S. Government Printing Office, 1949. 8+95 pp. 40 cents.

Precis de Mathematiques Economiques et Fiscales. By Henri Eyraud. Lyon, France, Faculté des Sciences, Lyon, 1948. 72 pp.

Modern Algebraische Geometrie. Die Idealtheoretischen Grundlagen. By Wolfgang Fröbner. Springer Verlag, 1949, 12+212 pp., 1949. $5.70.

CLUBS AND ALLIED ACTIVITIES

EDITED BY L. F. OLLMANN, Hofstra College

Send reports of all activities, such as club reports, special features, topics with references, student papers, and other material of interest to L. F. Ollmann, Hofstra College, Hempstead, New York.

CLUB REPORTS, 1948–49

White Mathematics Club, University of Kentucky

The *White Mathematics Club* met monthly during the year except for the months of September and December. Papers presented include:

Use of the analytical triangle in curve tracing, by Virginia Baskett

The use of complex numbers in linear network analysis, by J. C. Flack

Integral domains, by A. E. Foster

Jobs for mathematics majors, by T. K. Dyer.

A birthday party for Dr. H. H. Downing, Head of the Department of Mathematics and Astronomy, was held in November.

A social get-together was held at the end of each regular meeting and refreshments were served. The annual Club Picnic was held in May.

The officers elected were: President, Fay Hays; Vice-President, Elizabeth Napier; Secretary-Treasurer, Eugene Miller; Chairman of Publicity Committee, Franz Ross.

Zeno Club, Alfred University

The following talks were given at the bimonthly meetings of the *Zeno Club* of Alfred University:

The normal curve, by Mr. H. S. Graf

Multiple surfaces, by Dr. E. Rhodes

Kirkman's schoolgirl problem, by Mr. R. Beals

Pick a number, by Prof. V. Nevins

Differential equations of applied mathematics, by Mr. I. Miller

The mil system, by Mr. H. Munson

Interpolation errors, by Dr. E. Rhodes

The Lorentz transformation, by Prof. J. Freund

An introduction to the solution of indeterminate problems, by Prof. R. Polan

Non-euclidean geometry, by Mr. L. Shershoff.

The first annual mathematics contest for freshmen and sophomores, sponsored by the *Zeno Club*, was won by Mr. H. L. Stoll. Mr. H. Munson was chosen as the outstanding mathematics student of the year at Alfred University.

Officers elected for 1949–50 are: President, L. Shershoff; Vice-President, I. Miller; Secretary, J. Freund.

Ricci Mathematics Academy, Boston College

The following talks were presented to the *Ricci Mathematics Academy*, Boston College, during 1948–49:

The theory and use of the slide rule, by Thomas Colbert

The philosophy of mathematics, by Dr. Fakhri Maluf

Mathematics as the language of science, by Rev. J. A. Tobin, S.J.

On the Ricci Mathematical Journal, by Patrick Leonard and John McClay

Curves and determinants, by Dr. F. E. White

Fields open to the graduate in mathematics, by George Donaldson

The lighter side of mathematics, by Rev. J. F. X. Murphy, S.J.

The first Annual Mid-term Social and Dance, sponsored by the Academy, was held at the Hotel Commander in Cambridge.

Officers for the year of 1949–50 were: President, Anthony Minnichelli; Vice-President, John Monahan; Secretary, Anthony Lemos; Treasurer, Edmund Murphy; Moderator, Joseph Krebs.

Mathematics-Physics Club, College of Saint Teresa

In addition to a Christmas party and a Spring picnic, four meetings were held at which the following papers were presented by members of the club:

The atomic age and some of its social and moral implications

Popular astronomy

Time and its measurement

Science and its impact on the literary imagination, by Sister M. Emmanuel, O.S.F.

Other activities included a field trip to the Mayo Research Foundation, the Medical Science Building, and the Mayo Clinic in Rochester, Minnesota.

Officers for the year 1949–50 are: President, Mary Ellen Tighe; Vice-President, Rita Kulas; Secretary, Eleanor Wise; Treasurer, Nancy Tighe.

Mathematics Club, Carleton College

The *Mathematics Club* of Carleton College held monthly meetings which included a picnic, a Christmas party, and a program of mathematical recreations. The movie *A triple integral* was also shown. The talks presented during the year 1948–49 were:

Calculating machines, by Dr. K. May

Graphical representation of complex roots, by Prof. L. Beasley

Mathematical implications of Zeno's paradoxes, by Prof. M. Capek

Wire puzzles, by Prof. C. Hatfield

Review of "Flatland," by E. A. Abbott, by Miss Jeane M. Baldwin.

Officers for the year 1948–49 were: President, Joan Snapper; Vice-President, Kinsey Anderson; Secretary, Yolanda Stork; Faculty Advisor, Prof. Kenneth May.

NEWS AND NOTICES

EDITED BY EDITH R. SCHNECKENBURGER, University of Buffalo

Readers are invited to contribute to the general interest of this department by sending news items to Edith R. Schneckenburger, University of Buffalo, Buffalo 14, New York. Items must be submitted at least two months before publication can take place.

RESEARCH FELLOWSHIPS IN PSYCHOMETRICS

The Educational Testing Service is offering for 1950–51 its third series of research fellowships in psychometrics leading to the Ph.D. degree at Princeton University. Open to men who are acceptable to the Graduate School of the University, the two fellowships each carry a stipend of $2,375 a year and are normally renewable.

Fellows will be engaged in part-time research in the general area of psychological measurements at the offices of the Educational Testing Service and will, in addition, carry a normal program of studies in the Graduate School. Competence in mathematics and psychology is a prerequisite for obtaining these fellowships. Information and application blanks may be obtained from: Director of Psychometric Fellowship Program, Educational Testing Service, Box 592, Princeton, N. J.

PERSONAL ITEMS

Cornell University announces: Assistant Professor G. L. Walker of Purdue University has been appointed Visiting Assistant Professor of Mathematics; Dr. Bertram Yood has been promoted to an assistant professorship; Assistant Professor G. B. Robison of Sampson College has been appointed to a teaching fellowship.

Marietta College reports that Mr. R. E. Eberhard has been appointed to an instructorship and that Miss Margaret Bootz has resigned to continue graduate study at Ohio State University.

Oberlin College makes the following announcements: Acting Chairman E. P. Vance has been promoted to the position of Chairman of the Department of Mathematics; Assistant Professor R. W. Wagner has been promoted to an associate professorship; Dr. Bryant Tuckerman of Cornell University has been appointed to an assistant professorship.

At the University of Colorado: Professor A. J. Kempner has retired and has accepted a visiting professorship at Pomona College; Dr. A. B. Farnell of Princeton University has been appointed to an assistant professorship.

Wayne University announces: Assistant Professor Russell Ackoff has been transferred from the Department of Philosophy to the Department of Mathematics; Mr. Samuel Conte has returned from a leave of absence; Mr. Bertram Eisenstadt has been appointed Assistant Professor of Mathematics.

Dr. Milton Abramowitz of the Computation Laboratory of the National Bureau of Standards is now associated with the Numerical Mathematics Service of New York City.

Instructor H. P. Atkins, Jr., of the University of Rochester has been promoted to an assistant professorship.

Assistant Professor A. H. Bailey of Georgia Institute of Technology has been promoted to an associate professorship.

Dr. T. A. Bancroft, formerly director of the Statistical Laboratory of Alabama Polytechnic Institute, has accepted a position at the Statistical Laboratory, Iowa State College.

Lecturer J. D. Bankier of McMaster University has been promoted to an associate professorship.

Dr. W. E. Barnes of Cornell University has been appointed to an assistant professorship at the College of William and Mary.

Mr. D. Y. Barrer of Northwestern University has been promoted to an instructorship.

Assistant Professor I. L. Battin of Drew University has been promoted to an associate professorship.

Mr. A. V. Bauser, formerly engineer, Elliott Company, Ridgeway, Pennsylvania, is now teaching at J. W. Cooper High School, Shenandoah, Pennsylvania.

Associate Professor J. W. Beach of the New Mexico School of Mines has been appointed Assistant Professor of Mathematics at the University of New Mexico.

Mr. R. W. Beals has been appointed to an instructorship at Alfred University.

Assistant Professor S. Louise Beasley of Carleton College has been appointed to an assistant professorship at Lindenwood College.

Instructor R. F. Bell has been promoted to an assistant professorship at Eastern Washington College of Education.

Assistant Professor F. L. Celauro of Newark College of Engineering has been appointed Associate Professor and Head of the Department of Mathematics at Western State College of Colorado.

Dr. Sarvadaman Chowla of the Institute for Advanced Study has been appointed to an acting professorship at the University of Kansas.

Professor J. W. Clawson has been appointed Dean of Ursinus College.

Assistant Professor V. F. Cowling of Lehigh University has been appointed to an associate professorship at the University of Kentucky.

Instructor R. R. Croxton of the University of South Carolina has been promoted to the position of Adjunct Professor of Mathematics.

Associate Professor J. C. Currie of Alabama Polytechnic Institute has accepted an associate professorship at Georgia Institute of Technology.

Mr. J. E. Darraugh of the Consolidated Edison Company, New York City, has been appointed to an instructorship at Case Institute of Technology.

Professor D. C. Dearborn, Catawba College, has been appointed Dean.

Assistant Professor F. G. Dressel of Duke University has been promoted to an associate professorship.

Mr. W. M. Duke, chairman of Research Division, Cornell Aeronautical Laboratory, has been promoted to the position of Assistant Director.

Dean W. H. Durfee has been promoted to the position of Provost of the Colleges of the Seneca.

Reverend L. E. Ernsdorff of Loras College has been promoted from an instructorship to an associate professorship.

Professor F. A. Ficken of the University of Tennessee has been granted a leave of absence for the year 1949–50 and is spending the year at the Institute of Mathematics and Mechanics, New York University.

Dr. C. D. Firestone, Rutgers University, has been promoted to an assistant professorship.

Dr. M. M. Flood of the American Statistical Association has accepted a position as project officer with the Rand Corporation, Santa Monica.

Associate Professor C. B. Gass of Nebraska Wesleyan University is now Professor of Mathematics and Dean of Men.

Mr. B. T. Goldbeck, Jr., graduate student at Texas Christian University, has been appointed to an instructorship at the University of Wyoming.

Assistant Professor Parker Hamilton, formerly of Boston University, has accepted an appointment at Antioch College.

Mr. R. A. Harrison, formerly master at St. Mark's School, Southborough, Massachusetts, has been appointed Chairman of the Department of Mathematics of The Peddie School, Hightstown, New Jersey.

Professor M. A. Hill, Jr., of the University of North Carolina is now Associate Dean of the General College.

Miss Ruth I. Hoffman who has been teaching at North High School, Denver, is now Dean at Byers Jr. High School, Denver.

Mr. W. C. Hoffman, Iowa State College of Agriculture and Mechanic Arts, has accepted a position as assistant social scientist with the Rand Corporation, Santa Monica.

Assistant Professor C. H. Holton has been promoted to an associate professorship at Georgia Institute of Technology.

Associate Professor G. B. Huff of the University of Georgia has been promoted to a professorship.

Assistant Professor M. Gweneth Humphreys of Newcomb Memorial College, Tulane University, has been appointed to an associate professorship at Randolph-Macon Woman's College.

Dr. Solomon Hurwitz, City College of the City of New York, has been promoted to an assistant professorship.

Instructor B. C. Horne, Jr., Virginia Polytechnic Institute, has been promoted to an assistant professorship.

Professor J. B. Jeffries, Agricultural and Technical College of North Carolina, has accepted a position as assistant director with the Midway Technical School, New York City.

Associate Professor R. E. Johnson of Yale University has been appointed to an associate professorship at Smith College.

Mr. Winfield Keck of Lafayette College has been promoted to an assistant professorship.

Instructor W. J. Klimczak of the University of Rochester has been promoted to an assistant professorship.

Mr. J. A. LaRue, formerly teaching fellow at West Virginia University, has been appointed to an instructorship at Morris Harvey College.

Mrs. Sally I. Lipsey, teacher at Taft High School, New York City, has been appointed Lecturer at Hunter College.

Assistant Professor Ella Marth, Harris Teachers College, is now Associate Professor of Mathematics and Dean of Women.

Mr. A. L. Mayerson, Institute of Life Insurance, has accepted a position as actuarial assistant with the National Surety Corporation, New York City.

Associate Professor F. J. Murray of Columbia University has been promoted to a professorship.

Mr. J. D. Newburgh of the Massachusetts Institute of Technology has been appointed a member of the Institute for Advanced Study.

Miss Margaret Owchar, formerly instructor at Rockford College, has been appointed to an instructorship at the University of Minnesota.

Mr. C. L. Perry of the University of Michigan has been appointed to an assistant professorship at the University of Arkansas.

Professor W. G. Pollard has been appointed Executive Director of the Oak Ridge Institute of Nuclear Studies.

Associate Professor D. H. Porter is on leave of absence from Marion College and has a position as teaching fellow at Indiana University.

Dr. R. C. Prim of Princeton University has accepted a position as mathematician with the Bell Telephone Laboratories.

Assistant Professor A. L. Putnam of the University of Chicago has been appointed to a professorship at New York University.

Dr. H. J. Ryser of the Institute for Advanced Study has been appointed to an assistant professorship at Ohio State University.

Dr. W. R. Scott, University of Michigan, has received an appointment as assistant professor at the University of Kansas.

Assistant Professor W. T. Scott, Northwestern University, has been promoted to an associate professorship.

Mr. K. M. Siegel is now Research Associate at the Aeronautical Research Center, University of Michigan, Willow Run Airport, Ypsilanti.

Assistant Professor W. C. Taylor, University of Cincinnati, has accepted a position as mathematician with the Ballistics Laboratory, Aberdeen Proving Ground, Maryland.

Mr. E. F. Trombley, formerly assistant instructor at the University of Chicago, has been appointed to an instructorship at Illinois Institute of Technology.

Professor J. A. Ward, who has been serving as assistant head of the Department of Mathematics of the University of Georgia, has been appointed to a professorship at the University of Kentucky.

Professor K. L. Warren of Millsaps College has been appointed Associate Professor of Physics, Kent State University.

Dr. V. M. Wolontis of Harvard University has been appointed to an assistant professorship at the University of Kansas.

Instructor H. M. Zerbe, Hazleton Center of Pennsylvania State College, has been promoted to an assistant professorship.

Reverend J. E. Case, S.J., director of the Department of Mathematics of St. Louis University, died on August 5, 1949.

Dr. J. K. L. MacDonald of the Naval Ordnance Test Station, China Lake, died February 3, 1949.

Professor A. H. Mowbray of the University of California died on January 7, 1949.

THE MATHEMATICAL ASSOCIATION OF AMERICA

Official Reports and Communications

THE THIRTY-FIRST SUMMER MEETING OF THE ASSOCIATION

The thirty-first summer meeting of the Mathematical Association of America was held at the University of Colorado, Boulder, Colorado, on Monday and Tuesday, August 29–30, 1949 in conjunction with the summer meetings of the American Mathematical Society, the Institute of Mathematical Statistics, and the Econometric Society. A total of seven hundred and seventy adults were registered, including the following three hundred and thirty-five members of the Association:

V. W. Adkisson, University of Arkansas
O. W. Albert, University of Redlands
E. B. Allen, Rensselaer Polytechnic Institute
R. D. Anderson, University of Pennsylvania
R. V. Anderson, Colorado A & M College
R. V. Andree, University of Oklahoma
K. J. Arnold, University of Wisconsin
W. L. Ayres, Purdue University
Joshua Barlaz, Rutgers University
I. A. Barnett, University of Cincinnati
C. F. Barr, University of Wyoming
D. Y. Barrer, Northwestern University
D. L. Barrick, University of Colorado
H. G. H. Bartram, Cornell University
M. A. Basoco, University of Nebraska
P. T. Bateman, Institute for Advanced Study
J. W. Beach, University of New Mexico
J. E. Bearman, University of Minnesota
E. F. Beckenbach, University of California at Los Angeles
May M. Beenken, Immaculate Heart College
Alice K. Bell, Fresno State College
J. H. Bell, Michigan State College
B. C. Bellamy, Bellamy & Sons
Theodore Bennett, Marietta College
S. F. Bibb, Illinois Institute of Technology
F. C. Biesele, University of Utah
E. E. Blanche, United States Army
H. D. Block, Iowa State College
R. P. Boas, Mathematical Reviews
H. F. Bohnenblust, California Institute of Technology
A. W. Boldyreff, University of New Mexico
T. A. Botts, University of Virginia

J. W. Bradshaw, University of Michigan
H. E. Bray, Rice Institute
J. R. Britton, University of Colorado
J. C. Brixey, University of Oklahoma
B. K. Brown, James Millikin University
M. C. Brown, University of Kentucky
R. C. Buck, Brown University
C. E. Buell, Sandia Laboratory
P. B. Burcham, University of Missouri
Herbert Buseman, University of Southern California
J. H. Bushey, Hunter College
Jewell H. Bushey, Hunter College
C. H. Butler, Western Michigan College
S. S. Cairns, University of Illinois
W. D. Cairns, Oberlin College
J. W. Calkin, Rice Institute
E. A. Cameron, University of North Carolina
C. C. Camp, University of Nebraska
H. H. Campaigne, United States Navy
F. M. Carpenter, Colorado School of Mines
W. B. Carver, Cornell University
L. G. Chelius, Atomic Energy Commission
R. V. Churchill, University of Michigan
A. G. Clark, Colorado A & M College
Helen E. Clarkson, Creighton University
Nathaniel Coburn, University of Michigan
C. J. Coe, University of Michigan
Nancy Cole, Syracuse University
E. P. Coleman, Highland Falls, N. Y.
L. A. Colquitt, Texas Christian University
E. G. H. Comfort, Illinois Institute of Technology
J. A. Cooley, University of Tennessee
N. A. Court, University of Oklahoma
W. R. Cowell, Kansas State College
K. W. Crain, Purdue University
E. L. Crow, U. S. Naval Ordnance Test Station
J. H. Curtiss, National Bureau of Standards
J. A. Daum, Texas A & M College
P. H. Daus, University of California at Los Angeles
M. W. De Jonge, Purdue University
R. F. Deniston, Iowa State College
W. W. Denton, University of Arizona
R. P. Dilworth, California Institute of Technology
Bernard Dimsdale, Aberdeen Proving Ground
Roy Dubisch, Fresno State College
W. L. Duren, Tulane University
J. M. Earl, University of Omaha
E. D. Eaves, University of Tennessee
Paul Eberhart, Washburn University
P. D. Edwards, Ball State Teachers College
W. E. Ekman, University of South Dakota
Terrell Ellis, North Texas State College
Paul Erdös, University of Illinois
G. C. Evans, University of California
H. P. Evans, University of Wisconsin
Howard Eves, Oregon State College
G. M. Ewing, University of Missouri
A. B. Farnell, University of Colorado
William Feller, Cornell University
H. H. Ferns, University of Saskatchewan
F. N. Fisch, Colorado State College of Education
C. H. Fischer, University of Michigan
L. R. Ford, Illinois Institute of Technology
W. C. Foreman, University of Kansas
J. S. Frame, Michigan State College
Thornton C. Fry, Bell Telephone Laboratories
L. E. Fuller, University of Wisconsin
M. G. Gaba, University of Nebraska
H. M. Gehman, University of Buffalo
F. C. Gentry, Arizona State College
R. E. Gilman, Brown University
J. W. Givens, University of Tennessee
A. M. Gleason, Harvard University
Michael Golomb, Purdue University
R. N. Goss, Iowa State College
W. H. Gottschalk, University of Pennsylvania
Cornelius Gouwens, Iowa State College
J. W. Green, University of California at Los Angeles
R. E. Greenwood, University of Texas
Edison Greer, Kansas State College
F. L. Griffin, Reed College
V. G. Grove, Michigan State College
H. T. Guard, Colorado A & M College
D. F. Gunder, Cornell University
W. S. Gustin, Indiana University
Beatrice L. Hagen, Pennsylvania State College
Franklin Haimo, Washington University
P. R. Halmos, University of Chicago
O. H. Hamilton, Oklahoma A & M College
Clara L. Hancock, Virginia Junior College
J. R. Hanna, University of Wichita
W. R. Hanson, City College of San Francisco
Katharine E. Hazard, Rutgers University
E. A. Hazlewood, Texas Technological College
I. L. Hebel, Colorado School of Mines
G. A. Hedlund, Yale University
E. R. Heineman, Texas Technological College
E. D. Hellinger, Illinois Institute of Technology

Ann S. Henriques, University of Utah
Gertrude A. Herr, Iowa State College
J. G. Herriot, Stanford University
Fritz Herzog, Michigan State College
M. R. Hestenes, University of California at Los Angeles
T. H. Hildebrandt, University of Michigan
A. G. Hill, North Dakota Agricultural College
J. J. L. Hinrichsen, Iowa State College
P. G. Hoel, University of California at Los Angeles
D. L. Holl, Iowa State College
Carl Holtom, U.S.A.F. Institute of Technology
LeRoy Holubar, University of Colorado
R. E. Horton, Los Angeles City College
L. Aileen Hostinsky, Temple University
E. Marie Hove, Hofstra College
J. M. Howell, Los Angeles City College
H. K. Hughes, Purdue University
P. F. Hultquist, University of Colorado
M. Gweneth Humphreys, Randolph-Macon Woman's College
N. C. Hunsaker, Utah State College
Burrowes Hunt, University of Colorado
C. C. Hurd, International Business Machines Corp.
W. R. Hutcherson, University of Florida
C. A. Hutchinson, University of Colorado
H. H. Irwin, State College of Washington
T. W. Jackson, Jamestown College
B. W. Jones, University of Colorado
G. K. Kalisch, University of Minnesota
Irving Kaplansky, University of Chicago
Lois Karr, Lindenwood College
M. W. Keller, Purdue University
J. L. Kelley, University of California
Claribel Kendall, University of Colorado
P. W. Ketchum, University of Illinois
D. E. Kibbey, Syracuse University
E. C. Kiefer, James Millikin University
W. J. Kirkham, Oregon State College
J. R. Kline, University of Pennsylvania
H. L. Krall, Pennsylvania State College
Max Kramer, University of Illinois
G. M. Kuznets, University of California
O. E. Lancaster, United States Navy
Joseph Landin, University of Illinois
R. E. Langer, University of Wisconsin
H. D. Larsen, Albion College
C. G. Latimer, Emory University
D. H. Leavens, Colorado Springs, Colo.
W. G. Leavitt, University of Nebraska
Solomon Lefschetz, Princeton University
D. H. Lehmer, University of California
L. C. Leithold, Phoenix College
W. T. Lenser, University of Nebraska
W. J. LeVeque, University of Michigan
Harry Levy, University of Illinois
A. J. Lewis, University of Denver
C. F. Lewis, Kansas State College
F. A. Lewis, University of Alabama
S. B. Littauer, Columbia University
W. S. Loud, University of Minnesota
R. G. Lubben, University of Texas
Saunders MacLane, University of Chicago
M. L. Madison, Colorado A & M College
Morris Marden, University of Wisconsin
Anna Marm, Bethany College
D. C. B. Marsh, University of Arizona
M. H. Martin, University of Maryland
J. R. Mayor, University of Wisconsin
R. B. McClenon, Grinnell College
Eloise McCord, University of Wichita
Dorothy McCoy, Wayland College
N. H. McCoy, Smith College
L. H. McFarlan, University of Washington
E. J. McShane, University of Virginia
A. E. Meder, Jr., Rutgers University
A. S. Merrill, Montana State University
B. C. Meyer, University of Arizona
H. L. Meyer, Jr., University of Chicago
R. R. Middlemiss, University of Washington
R. A. Miller, University of Mississippi
T. W. Moore, U. S. Naval Academy
F. R. Morris, Fresno State College
Dorothy J. Morrow, Civil Aeronautics Administration
S. B. Myers, University of Michigan
W. K. Nelson, University of Colorado
Greta Neubauer, University of Wyoming
C. V. Newsom, University of the State of New York
Ivan Niven, University of Oregon
M. L. Norden, Johns Hopkins University
Ruth E. O'Donnell, Duquesne University
E. G. Olds, Carnegie Institute of Technology
Emma J. Olson, Kent State University
R. L. O'Quinn, Louisiana State University
J. C. Oxtoby, Bryn Mawr College
H. C. Peterson, University of Denver
R. P. Peterson, University of California at Los Angeles
T. S. Peterson, University of Oregon
A. D. Pierson, Jr. College of Kansas City

MARGARET M. PIHLBLAD, University of Kansas
GEORGE PIRANIAN, University of Michigan
J. C. POLLEY, Wabash College
FLORENCE E. POOL, University of Nebraska
G. B. PRICE, University of Kansas
A. L. PUTNAM, University of Chicago
E. D. RAINVILLE, University of Michigan
J. F. RANDOLPH, University of Rochester
RUTH B. RASMUSEN, Chicago City College
L. T. RATNER, Vanderbilt University
L. L. RAUCH, University of Michigan
C. B. READ, University of Wichita
M. O. READE, University of Michigan
O. H. RECHARD, University of Wyoming
O. W. RECHARD, Ohio State University
J. K. RECKZEH, New Jersey State Teachers College
MINA S. REES, United States Navy
P. K. REES, Louisiana State University
R. F. REEVES, Iowa State College
W. T. REID, Northwestern University
IRVING REINER, University of Illinois
J. G. RENNO, University of Wisconsin
C. N. REYNOLDS, West Virginia University
H. L. RICE, University of Omaha
F. A. RICKEY, Louisiana State University
P. R. RIDER, WASHINGTON University
L. G. RIGGS, Northwestern University
L. A. RINGENBERG, Eastern Illinois State College
D. D. RIPPE, University of Michigan
E. K. RITTER, University of Michigan
B. D. ROBERTS, New Mexico Highlands University
FRED ROBERTSON, Iowa State College
RAPHAEL M. ROBINSON, University of California
FLORENCE V. ROHDE, University of Kentucky
ARTHUR ROSENTHAL, Purdue University
J. B. ROSSER, Institute for Numerical Analysis
M. F. ROSSKOPF, Syracuse University
R. G. SANGER, Kansas State College
A. C. SCHAEFFER, Purdue University
EDITH R. SCHNECKENBURGER, University of Buffalo
K. C. SCHRAUT, University of Dayton
NATHAN SCHWID, University of Wyoming
E. J. SHAPIRO, Brooklyn College
H. C. SHAUB, Washington & Jefferson College
C. R. SHERER, Texas Christian University
L. W. SHERIDAN, College of St. Thomas
SISTER M. NICHOLAS ARNOLDY, Marymount College
SISTER M. PACHOMIA LARKEY, College of St. Teresa
SISTER M. TERESINE LEWIS, Fontbonne College
SISTER ROSE MARGARET COOK, Loretto Heights College
M. F. SMILEY, State University of Iowa
A. H. SMITH, Purdue University
F. C. SMITH, College of St. Thomas
G. W. SMITH, University of Kansas
H. L. SMITH, Louisiana State University
L. C. SNIVELY, University of Colorado
ELIZABETH S. SOKOLNIKOFF, University of Wisconsin
I. S. SOKOLNIKOFF, University of California at Los Angeles
T. H. SOUTHARD, Wayne University
F. W. SPARKS, Texas Technological College
E. J. SPECHT, Emmanuel Missionary College
VIVIAN E. SPENCER, U. S. Bureau of Census
C. E. SPRINGER, University of Oklahoma
K. H. STAHL, University of Colorado
L. W. STARK, Atlantic Christian College
B. M. STEWART, Michigan State College
RUTH W. STOKES, Syracuse University
E. B. STOUFFER, University of Kansas
C. J. STOWELL, McKendree College
A. G. SWANSON, Gustavus Adolphus College
OTTO SZASZ, University of Cincinnati
A. HELEN TAPPAN, Western College
A. H. TAUB, Institute for Advanced Study
V. B. TEMPLE, Louisiana College
H. P. THIELMAN, Iowa State College
J. M. THOMAS, Duke University
F. B. THOMPSON, University of California
R. M. THRALL, University of Michigan
LEONARD TORNHEIM, University of Michigan
E. P. TOVANI, University of Colorado
A. W. TUCKER, Princeton University
J. W. TUKEY, Princeton University
J. L. ULLMAN, University of Michigan
GILBERT ULMER, University of Kansas
R. S. UNDERWOOD, Texas Technological College
W. R. UTZ, University of Missouri
H. S. VANDIVER, University of Texas
V. J. VARINEAU, University of Wyoming
R. W. VEATCH, University of Tulsa
J. F. WAGNER, University of Colorado
EARL WALDEN, New Mexico College of A & M A
G. L. WALKER, Cornell University
R. J. WALKER, Cornell University
J. L. WALSH, Harvard University

Lillie C. Walters, University of Colorado
K. W. Wegner, Carleton College
A. L. WHITEMAN, University of Southern California
L. R. WILCOX, Illinois Institute of Technology
F. B. WILEY, Denison University
S. S. WILKS, Princeton University
R. L. WILSON, University of Tennessee
G. M. WING, Cornell University
R. M. WINGER, University of Washington
CLEMENT WINSTON, Department of Commerce
H. E. WOLFE, Indiana University
C. R. WYLIE, JR., University of Utah
P. M. YOUNG, Kansas State College
J. H. ZANT, Oklahoma A & M College

The Association held it sessions on Monday afternoon and Tuesday morning in Room 201W, Arts Building, with President R. E. Langer presiding. The Tuesday morning session was held jointly with the Institute of Mathematical Statistics and the Econometric Society. The Program Committee for the meeting consisted of S. S. Cairns, chairman, Wladimir Seidel, and C. R. Wylie, Jr.

FIRST SESSION OF THE ASSOCIATION

"What Has Happened to Algebraic Geometry?", by Professor R. J. Walker, Cornell University.

"The Problems Section of the Monthly," by Professor Howard Eves, Oregon State College.

"The Monte Carlo Method," by Dr. S. M. Ulam, Los Alamos Scientific Laboratory.

SECOND SESSION OF THE ASSOCIATION

Symposium: "Mathematical Training for Social Scientists," with Professor Jacob Marschak, University of Chicago as Chairman, and Professors R. L. Anderson, North Carolina State College, T. W. Anderson, Columbia University, G. C. Evans, University of California, F. L. Griffin, Reed College, Harold Gulliksen, Educational Testing Service, Harold Hotelling, University of North Carolina, William Jaffe, Northwestern University, and G. M. Kuznets, University of California, as participants.

At the conclusion of the symposium, it was voted that the officers of the Mathematical Association of America, the Institute of Mathematical Statistics, and the Econometric Society be requested to appoint a joint committee, in cooperation with other interested societies, to study the need for better mathematical training of social scientists and the ways and means of improving their mathematical preparation. It was suggested that this committee report at the next joint meeting of the three organizations.

MEETING OF THE BOARD OF GOVERNORS

The Board met on Monday evening in the Southwest Recreation Room of the Third Dormitory. Twenty-seven members of the Board were present.

Progress reports were received from several committees. No action was taken on the matter of proposed translations of publications of the Association into foreign languages, but the concensus was that everything possible should be done to encourage the development of mathematics in foreign countries, especially in South America.

The Board voted to authorize the printing of two new Carus Monographs. Number 9 is entitled: "The Theory of Algebraic Numbers" by Harry Pollard and number 10 is: "The Arithmetic Theory of Quadratic Forms" by B. W. Jones. The Executive and Finance Committees were empowered to act on the method of distribution of Monographs 9 and 10.

MEETING OF SECTION SECRETARIES

A meeting of secretaries of the Sections of the Association was held on Tuesday evening in the Southwest Lounge. Twenty-two of the twenty-five Sections were represented.

The general topic of the meeting was the proposed handbook of instructions for Section Secretaries. Many helpful suggestions were made concerning the conduct of section meetings. Representatives of several sections reported on special projects being carried out by their sections.

MEETINGS OF OTHER ORGANIZATIONS

The sessions of the American Mathematical Society began on Tuesday afternoon and continued through Friday afternoon. Professor G. A. Hedlund of Yale University delivered the Colloquium lectures on "Topological Dynamics." Professor B. Jessen of the University of Copenhagen spoke on "Some recent investigations in almost periodic functions" and Professor F. B. Jones of the University of Texas spoke on "Aposyndetic and non-aposyndetic continua."

The Institute of Mathematical Statistics held its sessions from Monday afternoon through Thursday afternoon.

Sessions of the Econometric Society began on Monday morning and continued through Friday morning.

From Monday through Thursday, the Ninth Summer Meeting of the National Council of Teachers of Mathematics was held at the University of Denver in Denver, Colorado. Many members of the Council and of the Association took advantage of this opportunity to attend the summer meetings of both organizations.

ARRANGEMENTS, ENTERTAINMENT AND RECREATION

The New Dormitory Group of the University of Colorado was available to all attending the meeting and to their families. Meals were served in the dining halls of the Third Dormitory.

On Tuesday afternoon a tea for the visiting mathematicians and their families was given at the President's house by the ladies of the mathematics departments.

On Wednesday afternoon an excursion was held along the Trail Ridge Road in Rocky Mountain National Park to the Museum and Store at Fall River Pass. Supper was eaten at Glacier Basin Camp Ground.

On Thursday evening a steak fry was held on Flagstaff Mountain. Afterwards the group gathered in the amphitheatre on the top of the mountain overlooking the University and the city of Boulder. Professor C. A. Hutchinson acted as master of ceremonies. Vice-President W. F. Dyde greeted the guests on behalf of the University of Colorado. Dr. C. V. Newsom responded on behalf of the four organizations. A telegram from Professor A. J. Kempner was read expressing his regrets at being unable to be present at the meeting and the group voted to send Professor Kempner a telegram of greetings and best wishes. Mr. A. B. Patterson, Episcopal student chaplain at the University of Colorado, led the singing and sang several solos.

At the conclusion of Dr. Newsom's talk, a motion was enthusiastically adopted expressing appreciation to the University of Colorado for the hospitality it had extended to the visitors, and thanking all those who had given so generously of their time in making the meetings such a noteworthy success. Special mention was made of the members of the departments of Mathematics and of Applied Mathematics, and of the Committee on Arrangements, and finally of C. A. (Hutch) Hutchinson, himself, "who had done everything from endorsing checks to ringing the dinner bell."

H. M. GEHMAN, *Secretary-Treasurer*

THE MARCH MEETING OF THE PACIFIC NORTHWEST SECTION

The third annual meeting of the Pacific Northwest Section of the Mathematical Association of America was held at Oregon State College, Corvallis, on Friday and Saturday, March 25–26, 1949.

Eighty-five persons attended, including the following fifty-seven members of the Association: Jessie V. Allhands, Tyler Allhands, H. A. Antosiewicz, R. A. Beaumont, R. F. Bell, S. E. Boselly, L. G. Butler, W. B. Caton, Richard Cebula, Paul Civin, C. L. Clark, C. M. Cramlet, J. H. Curtiss, Douglas Derry, W. W. Dolan, J. L. Ericksen, H. W. Eves, K. S. Ghent, E. G. Goman, F. L. Griffin, S. G. Hacker, V. E. Hoggatt, Jr., H. H. Irwin, S. A. Jennings, J. M. Kingston, W. J. Kirkham, Celia E. Klotz, M. S. Knebelman, J. C. R. Li, J. J, Livers, A. T. Lonseth, R. E. Lowney, C. F. Luther, L. H. McFarlan, W. E. Milne, A. F. Moursund, B. N. Moyls, D. C. Murdoch, Ivan Niven, Andrewa R. Noble, T. G. Ostrom, T. S. Peterson, A. R. Poole, H. F. Price, R. A. Rosenbaum, R. B. Saunders, W. G. Scobert, W. L. Shepherd, W. H. Simons, M. C. Stippes, W. M. Stone, D. B. Tillotson, J. R. Vatnsdal, G. A. Williams, L. B. Williams, R. M. Winger, F. E. Wood.

At the business meeting of the Section the following officers were elected for the year 1949–50: Chairman, R. M. Winger, University of Washington; Vice-Chairman, A. F. Moursund, University of Oregon; Secretary-Treasurer, S. G. Hacker, State College of Washington. The Section voted to accept the invitation of the University of Washington to hold the next annual meeting in Seattle in the spring of 1950.

The Friday afternoon session was opened by Professor W. E. Milne, Chairman of the Section, and Professor Moursund was invited by him to preside. By invitation the two papers by Professors Eves and Rosenbaum were assigned twenty minutes and each was followed by an interesting discussion. Subsequently, on motion of Professor Eves, a representative committee from this Section, with Professor Rosenbaum as chairman, was appointed to extend and implement the suggestions made in Professor Rosenbaum's paper; these suggestions are outlined below in the abstract of his paper. Professor C. M. Cramlet, University of Washington, delivered the annual invited hour address, the title being *On Algebraic and Differential Invariant Theory.*

Following the business meeting on Friday night Professor H. H. Irwin, State College of Washington, gave a report of the findings of the Committee on Improvements in Qualifications of Mathematics Teachers in Secondary Education in the Pacific Northwest. This committee was appointed last year by this Section, and it was asked to continue to serve for the year 1949–50. Professor Irwin reported on the results of three surveys conducted by his committee: (1) the requirements of Pacific Northwest institutions for a bachelor's degree in mathematics or for a degree in education with a major or minor in mathematics; (2) undergraduate mathematics courses considered by college instructors and by secondary school teachers to be helpful to the latter in their subsequent teaching; (3) the amount of college training in mathematics required by each of the forty-eight states in order for a secondary school teacher to qualify to teach mathematics in its accredited secondary schools.

On Saturday morning the newly elected Chairman, Professor Winger, invited Professors Griffin and Knebelman to preside. Dr. J. H. Curtiss of the National Applied Mathematics Laboratories, National Bureau of Standards, read an invited paper on *The Mathematical Programs of the National Bureau of Standards.* Dr. Curtiss spoke regarding the background, administration, and certain objectives of the Laboratories.

The first eight of the following papers, including Professor Cramlet's address, were presented at the Friday afternoon session. The other thirteen papers, were read on Saturday morning.

1. *A sum transformation*, by Professor W. M. Stone, Oregon State College.

A sum transformation, defined by

$$LF(kh) = h\sum_{n=0}^{\infty} F(nh)(1 + sh)^{-n-1}$$

where k is a positive integer and h and s are arbitrary complex parameters, reduces to the classical Laplace transformation when $|h|\to 0$, $kh\to t$. Various properties of the sum transformation were given, together with applications to the solution of difference equations, particularly those which arise by replacing differential operators by difference operators.

2. *Weighted trigonometric interpolation*, by Professor Paul Civin, University of Oregon.

For a continuous function $f(x)$, a comparison was made between the convergence behavior of the set of trigonometric polynomials of order n which take the values $f(x_n^i)$ at $x_n^i = 2\pi i/(2n+1)$, and the set which takes values

$$\alpha(n)[f(x_n^{i-1}) + f(x_n^{i+1})] + [1 - 2\alpha(n)]f(x_n^i)$$

at x_n^i.

3. *A parametric treatment of polar tangent curves*, by Professor R. M. Winger, University of Washington.

These curves have been studied by Stratton, utilizing mainly polar coordinates, in this MONTHLY, vol. 43 (1936), p. 398. Professor Winger derived a compact parametric representation, permitting the use of the technique for dealing with rational curves, and he obtained and employed to advantage the collineation groups which leave the curves invariant.

4. *The rank of a curve of real order 3 in 3-space*, by Professor Douglas Derry, University of British Columbia.

Denoting by C_3 a closed curve in real projective 3-space which is cut by a plane in at most three real points, for each point of which a tangent and osculating plane are defined, the author proved the rank of C_3 to be 4. An extension of the proof to a C_4 was indicated.

5. *An experiment in intellectual stimulation of gifted high school students*, by Professor L. B. Williams, Reed College.

During the past year members of the staff of the Science Division of Reed College have given monthly lectures on scientific subjects, followed by discussions, to a group of about sixty Portland high school students selected on the basis of scientific aptitude and interest. Professor Williams reported on the way in which students were selected and the lecture subjects, together with a partial evaluation of the response and the results.

6. *Invited Annual Address: On algebraic and differential invariant theory*, by Professor C. M. Cramlet, University of Washington.

Professor Cramlet traced the development of the theory of algebraic invariants from the early formal theory to the theory of group representations. It was explained that the latter provides an abstract basis on which the foundations of tensor algebra are securely built, algebraic invariant theory becoming an aspect of tensor algebra, and differential invariant theory a natural extension from the theory of algebraic forms to the theory of differential forms.

7. *Some unsolved "elementary" problems submitted to the Monthly*, by Professor H. W. Eves, Oregon State College.

The Elementary Problem Department of this MONTHLY made its first appearance in the October 1932 issue, and has since published over 850 proposals. It is interesting that among the first 800 of these problems there are eight for which no solutions have been received, thirteen for which only partial solutions have been received, and nine which contain unanswered related questions raised either by a solver or by the editor. The problem numbers of the proposals in the respective groups are: (1) E 9, E 28, E 518, E 534, E 570, E 585, E 604, E 774; (2) E 185, E 208, E 246, E 379, E 410, E 442, E 496, E 590, E 708, E 735, E 764, E 765, E 777; (3) E 400, E 401, E 476, E 644, E 696, E 724, E 747, E 750, E 791. Professor Eves' paper was devoted to a discussion of selected problems from this list.

8. *Discussion of a problem book of a particular type*, by Professor R. A. Rosenbaum, Reed College.

Professor Rosenbaum suggested that this Section might possibly be interested in the compilation of a small book of serious problems of definite mathematical content which could prove a challenge and a stimulating experience to undergraduates of ability in the course of their study of mathematics. The speaker gave several interesting illustrations of the types of problems which he has in mind.

9. *Note on formal integration*, by Professor F. L. Griffin, Reed College.

Professor Griffin called attention to systematic elementary procedures for integrating certain expressions of the types considered by H. F. MacNeish in this MONTHLY, vol. 56 (1949), p. 25.

10. *On the integral solutions of the Diophantine equations* $x^2-y^2=at^2$, $y^2-z^2=at^2$, by Mr. Joseph Irwin, Portland, Oregon, introduced by Professor Griffin, read by title.

11. *On the number of primitive* λ*-roots mod* m, by Professor D. C. Murdoch, University of British Columbia.

Professor Murdoch stated that in an unpublished paper H. Griffin and B. Sussman have solved the problem of finding the number of primitive λ-roots modulo a composite integer m, and he gave the following alternative formula obtained by group theoretic methods. If $m=p_0^{e_0}p_1^{e_1}\cdots p_r^{e_r}$, where $p_0=2$ and $p_1, \cdots, p_r$ are distinct odd primes, the number of primitive λ roots mod m is

$$\phi[\phi(m)]\cdot\prod_{i=0}^{r}\left[(1-q_i^{-\nu_i})/(1-q_i^{-1})\right]$$

where the product extends over all prime divisors q_i of $\phi(m)$, $q_0=2$, and the numbers ν_i are defined as follows: (a) if $e_0\leqq 2$ and $q_i^{\beta_i}$ is the highest power of q_i which divides any of the numbers $\phi(p_0^{e_0})$, $\phi(p_1^{e_1}), \cdots, \phi(p_r^{e_r})$, then ν_i is the number of these numbers which are divisible by $q_i^{\beta_i}$; (b) if $e_0>2$ and $q_i^{\beta_i}$ is the highest power of q_i which divides any one of the numbers 2, $\frac{1}{2}\phi(p_0^{e_0})$, $\phi(p_1^{e_1}), \cdots$, $\phi(p_r^{e_r})$, then ν_i is the number of these numbers which are divisible by q_i.

12. *Hilbert's geometric postulates as theorems*, by Miss Patricia M. Cowan, Reed College, introduced by Professor Griffin.

Miss Cowan, a student at Reed College, discussed the possible deduction of Hilbert's postulates for geometry from the Pieri postulates. Mimeographed copies of the Pieri definitions and postulates were distributed in order to facilitate the discussion.

13. *Numerical treatment of the Laplacian operator*, by Professor W. E. Milne, Oregon State College and The Institute for Numerical Analysis.

Given a plane harmonic function $U(x, y)$ and its values U_i at the nodes of a square mesh, formulas were obtained, with bounds for the error, for the approximate representation of the first and second order partial derivatives of $U(x, y)$, for interpolation on curvilinear boundaries, and for the Laplacian operator. These formulas were expressed in terms of linear combinations of the U_i at a suitable pattern of nodal points.

14. *Some elementary problems in probability*, by Professor Ivan Niven, University of Oregon.

Several examples were given to show that books on college algebra could use problems on probability of the continuous type as well as the usual problems of the discrete type. For example, it might be asked, what is the probability that the distance between two points taken at random on the boundary of a unit square be greater than some constant which could either be specified numerically or left as a parameter.

15. *The mathematical programs of the National Bureau of Standards*, by Dr. J. H. Curtiss, National Applied Mathematics Laboratories, National Bureau of Standards.

16. *Rigidity of Lennes polyhedra*, by Professor H. W. Eves, Oregon State College.

Cauchy has shown that every simple closed convex polyhedral surface is rigid. To date it is unknown whether there exist simple closed non-convex polyhedral surfaces which are non-rigid. A Lennes polyhedron is a simple closed polyhedral surface with triangular faces and such that the line segment joining any two vertices not on a common edge lies either wholly or partly outside the volume bounded by the surface. Such polyhedra are known to be non-convex. The author stated that from paper models of Lennes polyhedra one might be led to conjecture that such polyhedral surfaces are non-rigid, for they seem to possess a fair range of mobility. However it was the purpose of this paper to show that this apparent motion is accompanied by small distortions of the faces and that the polyhedra are really rigid.

17. *A recursion relation involving exponentials*, by Mr. Daniel Drew, Reed College, introduced by Professor Rosenbaum.

Mr. Drew, a student at Reed College, obtained a general solution of lim u_n, $u_n = a^{u_{n-1}}$, $u_0 = a_0$, and applied the result to the problem of lim v_n, $v_n = a^{n^{n-1}}$, $v_0 = a_0$.

18. *An application of Laplace integrals to Hankel functions*, by Professor W. B. Caton, State College of Washington.

Using the principle of rotation of the path of integration of a Laplace integral, together with Abelian theorems for these integrals, the author obtained representations and formulas of the Hankel functions valid in appropriate half-planes. In particular, the expressions for $H_\alpha^{(1)}(ze^{2\pi i})$ and $H_\alpha^{(2)}(ze^{2\pi i})$ involving Laplace integrals were exhibited.

19. *Sample size for Wilks' tolerance limits*, by Professors Z. W. Birnbaum, and H. S. Zuckerman, University of Washington, read by title.

20. *Some consequences of a well-known theorem on conics*, by Professor R. A. Rosenbaum, Reed College, and Mr. Joseph Rosenbaum, The Milford School.

This paper, which was concerned with a generalization of the Desargues involution theorem and a porism related to Steiner's theorem on chains of tangent circles, was read by title at the authors' generous suggestion in view of the limited time left available.

21. *The Canadian congress on applied mathematics, Vancouver, 1949*, by Professor S. A. Jennings, University of British Columbia.

Professor Jennings gave a brief account of the extensive scientific and social arrangements which have been made for the Congress, and extended a cordial invitation to the members of this section to attend.

S. G. HACKER, *Secretary*

THE APRIL MEETING OF THE SOUTHWESTERN SECTION

The ninth annual meeting of the Southwestern Section of the Mathematical Association of America was held at New Mexico College of A. & M. A., State College, New Mexico, April 29 and 30, 1949. Professor Earl Walden, chairman of the Section, presided.

The attendance was forty-one including the following twenty-three members of the Association: J. W. Beach, A. W. Boldyreff, C. R. Buell, J. H. Butchart, Louise H. Chin, W. W. Denton, Ruth A. Fish, F. C. Gentry, R. F. Graesser, Rose A. Grundman, W. P. Heinzman, R. C. Hildner, P. F. Hultquist, H. D. Huskey, Brother Cyprian Luke, D. C. B. Marsh, E. J. Purcell, B. D. Roberts, H. P. Rogers, Earl Walden, R. L. Westhafer, Charles Wexler, and D. L. Webb.

The first forenoon was devoted to a trip to the White Sands Rocket Proving Grounds, where exhibits, lectures, and motion pictures instructed the group. A banquet at the Student Union Building was followed by an address, *Automatical Digital Computing*, by Dr. H. D. Huskey, Chief of the Machine Development Unit, Institute of Numerical Analysis, University of California at Los Angeles. This lecture was a joint presentation of the Association and the Physical Science Laboratory of the host college, Dr. George Gardiner, director.

At the business session the following officers were elected: Chairman, E. J. Purcell, University of Arizona; Vice-Chairman, A. W. Boldyreff, University of New Mexico; Visiting Lecturer, J. H. Butchart, Arizona State College. It was voted by the Section, upon the request of the delegation from the Texas School of Mines, El Paso, Texas, to ask the Board of Governors to change the territorial limits of the Southwestern Section to include El Paso, Texas. It was voted to invite the members at Lubbock, Texas to join in a similar request.

The program consisted of the following papers:

1. *On the definition of an analytical function of a complex variable*, by Professor A. W. Boldyreff, University of New Mexico.

2. *Systems of lines and planes on the quadric hypersurfaces of four and five dimensions*, by Ruth A. Fish, University of Arizona.

This was an expository treatment of systems of flat space on quadric primals in projective spaces of four and five dimensions.

3. *Postulates of projective algebras*, by Louise H. Chin, University of Arizona.

4. *Some characteristics of the altitude quadric*, by Professor J. H. Butchart, Arizona State College.

The speaker discussed some properties of the hyperboloid of one sheet of which the altitudes of a tetrahedron belong to one regulus. In particular, the Monge point is the center of the altitude quadric, and the plane section normal to any ruling is an equilateral hyperbola.

5. *A Cremona involution in n-dimensional space*, by Professor E. J. Purcell, University of Arizona.

6. *Grading and the fluctuations of sampling*, by Professor R. F. Graesser, University of Arizona.

A population of students with given percentages of A, B, C, D, and E students is assumed. From this population classes are drawn as random samples. The fluctuations in their grade distribution was exhibited.

7. *An application of matrix methods to a recent postulate system for m-valued functional calculi*, by Professor D. L. Webb, University of Arizona.

Professor Webb discussed an application of row matrices to the postulate system of J. B. Rosser and A. R. Turquette (*Journal of Symbolic Logic*, vol. 13, pp. 177–192). He compared the properties of the operators of the 2-valued and the m-valued cases.

8. *Further analysis of Fermat's congruence for composite moduli*, by Professor A. W. Boldyreff, University of New Mexico.

A method of finding all values of x satisfying the congruence $x^{n-1}\equiv 1 \pmod{n}$, when n is a product of two odd primes was discussed.

9. *Interpretation of singular solutions of ordinary differential equations of the first order by geometric means*, by Professor R. L. Westhafer, New Mexico College of A. & M. A.

Using the definition of singular and regular line element of the differential equation $F(x, y, p)=0$ as given by E. Kamke, in which the locus in 3-space of $F(x, y, z)=0$ is considered, the types of loci which give rise to singular line elements are seen to be isolated points and curves and their limit points, three dimensional regions of points, intersections of surfaces, and boundary points of the projections of the locus on the xy-plane.

10. *On certain analogies between measure and category*, by Manfred Fliess, New Mexico College of A. & M. A., introduced by the Secretary.

11. *On the three dimensional distribution of a bomb*, by M. S. Hendrickson, University of New Mexico, read by R. C. Hildner.

12. *A problem in maximum range for rockets*, by Keith Guard, New Mexico College of A. & M. A., introduced by the Secretary.

13. *Results of placement tests for sectioning college algebra*, by H. P. Rogers, University of New Mexico.

14. *On the high school training in mathematics of our freshmen*, by Professor Earl Walden, New Mexico College of A. & M. A.

B. D. Roberts, *Secretary*

THE APRIL MEETING OF THE METROPOLITAN NEW YORK SECTION

The eighth annual meeting of the Metropolitan New York Section of the Mathematical Association of America was held at Brooklyn College, Brooklyn, New York, on Saturday, April 9, 1949. Professor T. F. Cope, Collegiate Vice-Chairman of the Section presided at the morning session, and Professor R. A. Johnson, Chairman of the Section, presided at the afternoon session.

One hundred and twenty-five persons attended the sessions, including the following seventy-three members of the Association: Brother Bernard Alfred, R. G. Archibald, H. C. Ayres, Frances E. Baker, Samuel Borofsky, C. B. Boyer, A. D. Bradley, Benjamin Braverman, Paul Brock, A. B. Brown, Jewell Hughes Bushey, Hobart Bushey, Margaret C. Byrne, John Clark, T. F. Cope, J. E. Darraugh, J. G. Deutsch, I. A. Dodes, J. N. Eastham, J. E. Eaton, Samuel Eilenberg, Carolyn Eisele, J. M. Feld, Edward Fleisher, R. M. Foster, Marion C. Gray, Harriet Griffin, George Grossman, G. C. Helme, T. R. Humphreys, Solomon Hurwitz, L. C. Hutchinson, R. A. Johnson, Aida Kalish, L. S. Kennison, H. S. Kieval, M. S. Klamkin, Edna Kramer-Lassar, A. W. Landers, J. A. Larrivee, Nathan Lazar, A. A. LePori, M. E. Levenson, Emanuel Levine, May H. Maria, F. H. Miller, L. T. Moore, A. J. Mortola, D. S. Nathan, C. V. Newsom, M. A. Nordgaard, P. B. Norman, Walter Prenowitz, James Quinn, R. M. Reed, Moses Richardson, G. J. Ross, S. G. Roth, C. T. Salkind, Arthur Schack, Harry Schor, Aaron Shapiro, Edward Shapiro, James Singer, F. E. Smith, E. R. Stabler, Mildred M. Sullivan, Nelly Ullman, Israel Wallach, Alan Wayne, J. M. Wolfe, Margaret Y. Woodbridge, H. J. Zimmerberg.

The officers elected at the business meeting were: Chairman, B. P. Gill, The City College of the College of the City of New York; Collegiate Vice-Chairman, L. F. Ollman, Hofstra College; High School Vice-Chairman, Alan Wayne, Brooklyn High School of Automotive Trades; Secretary, James Singer, Brooklyn College; Treasurer, Aaron Shapiro, Midwood High School. The ninth annual meeting will be held in the Spring of 1950.

The following papers were presented:

1. *Address of welcome*, by Dr. W. R. Gaede, Dean of Faculty, Brooklyn College.

2. *Generalizations of the law of cosines*, by Professor L. W. Cohen, Queens College (introduced by Professor T. F. Cope).

3. *Interference patterns in the teaching of mathematics*, by Dr. Nathan Lazar, Bureau of Reference, Research and Statistics, Board of Education, New York City.

Experienced teachers of mathematics will not let a homework assignment consist of one type of exercise only, but will choose examples of each of several types. Nevertheless it is a common practice when introducing a topic to present only one aspect and to give almost exclusive drill on examples illustrating that new topic before any other topic is presented, no matter how closely related these topics may be.

This writer claims that the repetition of even one exercise without any significant variation in its pattern tends to encourage the student to perform mathematical operations without insight and understanding, and to make adventitious and unjustifiable inductions from the "model example" worked out. Further repetition of the same pattern will encourage the student to believe in the correctness of his procedure, and make it harder for him to adjust himself to a new type of the same pattern where his ad hoc hypothesis will not work.

It is therefore recommended that the successive presentation of examples illustrating the same mathematical concept be so varied as to prevent the formation of undesirable associations. This approach may not yield the immediate feeling of success that the traditional one gives. It may even require at first more time and more careful preparation than one is accustomed to. This effort will, however, be justified by the end result—a real understanding of the nature of mathematical operations, of the purposes underlying them, and of the mathematical laws governing them.

4. *The indebtedness of Greek to Babylonian exact sciences*, by Professor O. Neugebauer, Brown University (introduced by Professor T. F. Cope).

In 1928 J. K. Fotheringham published in the *Monthly Notices of the Royal Astronomical Society* an article on *The Indebtedness of Greek to Chaldean Astronomy*. In discussing this subject twenty years later, we can add Babylonian mathematics to the comparison. The vast increase of material has also considerably contributed to the complication of the problem and to the realization that there are huge gaps in our records. Greek and demotic papyri show that Babylonian arithmetical methods were used simultaneously with the geometrical astronomical models of Hipparchus and Ptolemy. Babylonian algebra developed methods which are paralleled in Greek "geometrical algebra." Babylonian number theory shows a development previously assumed to be "Pythagorean." The "Pythagorean" theorem was used a thousand years before Pythagoras. Yet it is very difficult to indicate the process by which these discoveries were transmitted to the Greeks. Theoretical astronomy still remains the only field where we are able to see some points of direct contact between Greek and Babylonian science.

5. *Some educational trends in New York State and their significance to the mathematicians*, by Dr. C. V. Newsom, Assistant Commissioner for Higher Education, University of State of New York.

This paper discussed some of the trends in education on all levels in New York State. Particular attention was given to the new program for the training of elementary and secondary teachers, the development of the new syllabus in mathematics for the secondary level, and the expanded program anticipated by the State in the field of higher education. The report contained many personal observations as the result of the author's actual visitation of institutions.

JAMES SINGER, *Secretary*

CALENDAR OF FUTURE MEETINGS

Thirty-third Annual Meeting, New York City, December 30, 1949.

International Congress of Mathematicians, Cambridge, Massachusetts, August 30–September 6, 1950.

The following is a list of the Sections of the Association with dates of future meetings so far as they have been reported to the Secretary.

ALLEGHENY MOUNTAIN

ILLINOIS, Southern Illinois University, Carbondale, May 12–13, 1950.

INDIANA, Wabash College, Crawfordsville, April 29, 1950.

IOWA, State University of Iowa, Iowa City, April 21–22, 1950.

KANSAS, Spring, 1950.

KENTUCKY, University of Kentucky, Lexington, April 29, 1950.

LOUISIANA-MISSISSIPPI, Centenary College, Shreveport, Louisiana, Spring, 1950.

MARYLAND-DISTRICT OF COLUMBIA-VIRGINIA, Fall, 1949.

METROPOLITAN NEW YORK, Spring, 1950.

MICHIGAN, March, 1950.

MINNESOTA, Macalaster College, St. Paul, May 6, 1950.

MISSOURI, Spring, 1950.

NEBRASKA, Nebraska Wesleyan University, Lincoln, May 6, 1950.

NORTHERN CALIFORNIA, Berkeley, January 28, 1950.

OHIO, Denison University, Granville, April 22, 1950.

OKLAHOMA

PACIFIC NORTHWEST, University of Washington, Seattle, June, 1950.

PHILADELPHIA, Haverford College, November 26, 1949.

ROCKY MOUNTAIN, University of Denver, April, 1950.

SOUTHEASTERN, University of Florida, Gainesville, March, 1950.

SOUTHERN CALIFORNIA, Immaculate Heart College, Hollywood, March 11, 1950.

SOUTHWESTERN, Spring, 1950.

TEXAS, Abilene, Spring, 1950.

UPPER NEW YORK STATE, Syracuse University, Spring, 1950.

WISCONSIN, Marquette University, Milwaukee, May, 1950.

GEORGE BANTA PUBLISHING COMPANY, MENASHA, WISCONSIN

10

M

THE AMERICAN MATHEMATICAL MONTHLY

THE OFFICIAL JOURNAL OF
THE MATHEMATICAL ASSOCIATION OF AMERICA, INC.

VOLUME 56 — NUMBER 10, Part 1

CONTENTS

DECEMBER 1949